To address this short coming we instead consider derivations acting on a stalk. Recall the definition of sheaves, stalks, and germs given on *pages 1026-1027*. Then given any C^r manifold M (with $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$) and any $0 \leq k \leq r$, we consider the sheaf $\mathcal{F} : \mathbf{Op}(M) \to \mathbf{CAlg}_{\mathbb{R}}$ such that $\mathcal{F}(U) = C^k(U, \mathbb{R})$ and for any open sets $V \subseteq U \subseteq M$ we have that $r_V^U(f) = f|_V$.

Here $C^0(U, \mathbb{R}) \coloneqq C(U, \mathbb{R})$.

For any $p \in M$ we denote the stalk of $\mathcal{F}$ at p to be the set $C_p^k(M)$. Also, given any open set U and C^k map $f : U \to \mathbb{R}$, we typically denote their germ as either $[(U, f)]$ or $[f]$.

Given any germ $[f] \in C_p^k(M)$, it is well-defined to evaluate $f(p)$.

This is because if $[(U, f)] = [(V, g)]$ then there is some open set $W \subseteq U \cap V$ containing p such that $f|_W = g|_W$. But in turn we know that $f(p) = g(p)$.

Now, we say a derivation on $C_p^k(M)$ is an $\mathbb{R}$-linear map $v : C_p^k(M) \to \mathbb{R}$ which also satisfies that if $[f]$ and $[g]$ are germs then $v([f][g]) = f(p)v([g]) + g(p)v([f])$. The set of all derivations on $C_p^k(M)$ is easily checked to be a real vector space and we denote that vector space by $\mathcal{D}_p(M)$.

Let M be an n-dimensional C^r manifold (where $r \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$). Then let $U \subseteq M$ be an open set, let $p \in M$, and consider any smooth chart (V, φ) such that $p \in V \subseteq U$. Note that φ defines local coordinates $(x_1, \ldots, x_n)$ on V. I'll also write $\varphi(p) = (x_1(p), \ldots, x_n(p))$.

Lemma 1: If $f \in C^k(U, \mathbb{R})$ (where $k \geq 1$) and $f(p) = 0$ then there is some open subset $W \subseteq V$ containing p such that:

$$f|_W = \sum_{i=1}^{n} (x_i - x_i(p)) h_i,$$

where $h_i \in C^{k-1}(W, \mathbb{R})$ for all i.

Proof:

To start off, if $\varphi(V)$ is not open in $\mathbb{R}^n$, then by definition $f \circ \varphi^{-1}$ can be extended to being a C^k function $\widetilde{f}$ from an open set $\widetilde{V} \subseteq \mathbb{R}^n$.

Next, let $0 < r$ be a real number such that $\overline{B_r(\varphi(p))} \subseteq \widetilde{V}$. Then for any $x = (x_1, \ldots, x_n) \in \overline{B_r(\varphi(p))}$ we can write:

$$\widetilde{f}(x) = \sum_{i=1}^{n} (x_i - x_i(p)) \int_0^1 \frac{\partial}{\partial x_i} \widetilde{f}(\varphi(p) + s(x - \varphi(p))) \mathrm{d}s$$

By Leibniz's rule, we can conclude that $\widetilde{h}_i : B_r(\varphi(p)) \to \mathbb{R}$ given by:

$$\widetilde{h}_i(x) = \int_0^1 \frac{\partial}{\partial x_i} \widetilde{f}(\varphi(p) + s(x - \varphi(p))) \mathrm{d}s$$

is a C^{k-1} map. And now we let $W = \varphi^{-1}(\varphi(V) \cap B_r(\varphi(p)))$ and $h_i : W \to \mathbb{R}$ be defined by $h_i = \widetilde{h}_i \circ \varphi$. And then each h_i will be a C^{k-1} map and $f|_W = \sum_{i=1}^{n} (x_i - x_i(p)) h_i$. ■

Now let (V, φ) be the smooth chart from before and suppose $f \in C^1(U, \mathbb{R})$ with $f(p) = 0$. If there exists an open set $W \subseteq U \cap V$ such that $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ where each $h_i \in C^0(W)$ and $h_i(\varphi(p)) = 0$ for each i, we say f vanishes to first order at p.

Lemma 2: Let $W, W' \subseteq U \cap V$ both be open sets containing p and then suppose $h_1, \ldots, h_n \in C^0(W, \mathbb{R})$ and $\overline{h_1}, \ldots, \overline{h_n} \in C^0(W', \mathbb{R})$ satisfy that:

$$f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i \text{ and } f|_{W'} = \sum_{i=1}^n (x_i - x_i(p))\overline{h_i}.$$

Then $h_i(p) = \overline{h_i}(p)$ for all i.

Proof:

By considering the set $W \cap W'$ we can assume without loss of generality that $W = W'$. Also, we can assume without loss of generality that W is an open set in $\mathbb{R}^n_{\geq 0}$, H^n, or $\mathbb{R}^n$ since $\overline{h_i}(p) = h_i(p) = 0$ iff $(\overline{h_i} \circ \varphi^{-1})(\varphi(p)) = (h_i \circ \varphi^{-1})(\varphi(p)) = 0$. Finally, note that $f = \sum_{i=1}^n (x_i - x_i(p))h_i = \sum_{i=1}^n (x_i - x_i(p))\overline{h}_i$ means that:

$$0 \equiv \sum_{i=1}^n (x_i - x_i(p)) \cdot (h_i - \overline{h_i}).$$

Thus, we will be done once we show that if $0 \equiv \sum_{i=1}^n (x_i - x_i(p))g_i$ where $g_i \in C^0(W, \mathbb{R})$ then $g_i(x(p)) = 0$ for all i (where $x(p) = (x_1(p), \ldots, x_n(p))$).

Fortunately, consider any $s > 0$ such that $x(p) + se_i \in W$. (Here $e_1, \ldots, e_n$ is the standard basis for $\mathbb{R}^n$). Then $0 = \int_0^s 0 \mathrm{d}t = \int_0^s t g_i(x(p) + te_i)\mathrm{d}t$. But now if $g_i(x(p)) \neq 0$ then by continuity we know there is some $s > 0$ and $\varepsilon > 0$ such that $|g_i(x(0) + te_i)| > \varepsilon$ when $0 \leq t \leq s$. In turn, we have a contradiction as $|\int_0^s t g_i(x(p) + te_i)\mathrm{d}t| \geq s\varepsilon$ means that $0 = \int_0^s t g_i(x(p) + te_i)\mathrm{d}t \neq 0$. ■

More specifically, what the prior lemma shows is that if $f \in C^1(U, \mathbb{R})$ with $f(p) = 0$ and $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ for some open set $W \subseteq U \cap V$ containing p and some C^0 maps $h_1, \ldots, h_n$ then:

$$h_i(p) = \frac{\partial}{\partial x_i}\Big|_p f \text{ for each } i.$$

After all, the above equation holds for the functions $h_1, \ldots, h_n$ constructed in lemma 1, and lemma 2 then says that all other possible $h_1, \ldots, h_n$ must agree with the functions constructed in lemma 1. As a result:

$$f \text{ vanishes to first order iff } \frac{\partial}{\partial x_i}\Big|_p f = 0 \text{ for all } i \text{ iff } v(f) = 0 \text{ for all } v \in T_pU$$

This leads naturally to another result:

Lemma 3: Now suppose (V', ψ) is another smooth chart containing p. We'll let ψ define the local coordinates $(y_1, \ldots, y_n)$ on V'. Also I'll write $\psi(p) = y(p) = (y_1(p), \ldots, y_n(p))$. If $W \subseteq U \cap V$ is an open set containing p and there exists functions $h_i \in C^0(W, \mathbb{R})$ such that $f|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$ and $h_i(p) = 0$ for each i, then for any open set $\Omega \subseteq U \cap V'$ containing p as well as functions $g_i \in C^0(\Omega, \mathbb{R})$ such that $f|_\Omega = \sum_{i=1}^n (y_i - y_i(p))g_i$, we know that $g_i(p) = 0$ for each i.

Proof:
We can conclude via lemmas 1 and 2 that $g_i(p) = \frac{\partial}{\partial y_i}\Big|_p f$ for each i. But now note from *page 1052* that each linear operator $\frac{\partial}{\partial y_i}\Big|_p$ is a linear combination of the operators $\frac{\partial}{\partial x_j}\Big|_p$. And since $\frac{\partial}{\partial x_j}\Big|_p f = h_j(p) = 0$ for each j by assumption, we can conclude $g_i(p) = 0$ for each i. ■

As a result of the prior lemmas, we can say that f vanishing to first order at p is independent of our chosen local coordinates or the particular functions in our expression:

$$f|_W = \sum_{i=1}^{n}(x_i - x_i(p))h_i.$$

Also as is hopefully obvious, if $[(U, f)] = [(\widetilde{U}, \widetilde{f})]$ in $C^1_p(M)$ and f vanishes to first order at p then so does $\widetilde{f}$. So, we can talk about a germ in $C^1_p(M)$ vanishing to first order.

Now here's the new and improved definition of the tangent plane of p. For now I'll denote my old construction as T_pM. Meanwhile, I'll define a new construction by:

$$\mathcal{T}_pM \coloneqq \{v \in \mathcal{D}_p(M) : v([f]) = 0 \text{ if } [f] \text{ vanishes to first order}\}.$$

Clearly $\mathcal{T}_pM$ is a vector subspace of $\mathcal{D}_p(M)$.

Theorem: There is a bijective linear map $\Phi : T_pM \to \mathcal{T}_pM$ such that $(\Phi(v))([f]) \coloneqq v(f)$ for all $f \in C^k(M, \mathbb{R})$.

Proof:
Suppose $[(U, f)]$ is any germ and note that by multiplying on a C^r bump function, we can find a function $g \in C^1(M, \mathbb{R})$ such that $[(M, g)] = [(U, f)]$. So, for any $v \in T_pM$ we let $(\Phi(v))([f]) \coloneqq v(g)$.

To see that Φ is well-defined, note that if $[(M, g)] = [(U, f)] = [(M, \widetilde{g})]$ then $g = f = \widetilde{g}$ on a neighborhood of p. So by proposition 3.8 on *page 1033* we know that $v(g) = v(\widetilde{g})$.

Showing that $\Phi(v) \in \mathcal{D}_p(M)$ is then easy. Furthermore, if $[(U, f)]$ vanishes to first order then we know for any $g \in C^k(M, \mathbb{R})$ with $[(M, g)] = [(U, f)]$ that $v(g) = 0$. So, $\Phi(v) \in \mathcal{T}_pM$. Finally, it is easy to verify that $\Phi(c_1v_1 + c_2v_2) = c_1\Phi(v_1) + c_2\Phi(v_2)$ for all $c_1, c_2 \in \mathbb{R}$ and $v_1, v_2 \in T_pM$. So, Φ is a linear map from T_pM to $\mathcal{T}_pM$.

To show that Φ is an isomorphism, let (U, φ) be a smooth chart containing p and defining local coordiantes $\varphi(m) = x(m) = (x_1(m), \ldots, x_n(m))$. Then recall $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p$ is a basis for T_pM. Meanwhile, we can define a basis for $\mathcal{T}_pM$ as follows:

Let $[(V, f)] \in C^1_p(M)$. By lemma 1 from a few pages ago can find an open subset $W \subseteq V \cap U$ as well as functions $h_i \in C^0(W, \mathbb{R})$ such that:

$$f|_W = f(p) + (x_i - x_i(p))h_i.$$

And now we define $\omega_i([f]) \coloneqq h_i(p)$. This is well-defined because if $[(V,f)] = [(\widetilde{V},\widetilde{f})]$ then we know there is some open set $\Omega \subseteq V \cap \widetilde{V}$ such that $f|_\Omega = \widetilde{f}|_\Omega$. And now upon writing that $f|_W = f(p) + \sum_{i=1}^n (x_i - x_i(p))h_i$ and $\widetilde{f}|_{\widetilde{W}} = f(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$, we know that:
$(f - f(p))|_{\Omega\cap\widetilde{W}} = \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$ and $(f - f(p))|_W = \sum_{i=1}^n (x_i - x_i(p))h_i$.

So by lemma 2 from before, we know that $h_i(p) = \widetilde{h}_i(p)$ for each i. And this proves that $\omega_i([f]) = \omega_i([\widetilde{f}])$.

Showing that $\omega_i \in \mathcal{T}_pM$ is now easy.

To show one necessary algebraic property of ω_i suppose $[f],[g] \in C_p^k(M)$ and write:
$f|_W = f(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$ and $g|_{\widetilde{W}} = g(p) + \sum_{i=1}^n (x_i - x_i(p))\widetilde{h}_i$.

Then after regrouping terms we can see that:
$$(fg)|_{W\cap\widetilde{W}} = (fg)(p) + \sum_{i=1}^n (x_i - x_i(p)) \cdot \left(h_i g + \widetilde{h}_i f - (x_i - x_i(p))h_i\widetilde{h}_i\right)$$

So, $\omega_i([f][g]) = \omega_i([f])g(p) + \omega_i([f])f(p) - 0$.

I also claim $\omega_1,\ldots,\omega_n$ span $\mathcal{T}_pM$. After all, let $\nu \in \mathcal{T}_pM$ be an arbitrary derivation.
We can consider the germ $[(U,f_i)]$ such that $f_i(m) = x_i(m) - x_i(p)$.
And then we let $c_i = \nu([f_i])$. Now for any germ $[f] \in C_p^k$, upon writing $f = f(p) + \sum_{i=1}^n (x_i - x_i(p))h_i$ we know that:
$f - f(p) - \sum_{i=1}^n (x_i - x_i(p))h_i(p)$ vanishes to first order at p.

So:
$$\begin{aligned} 0 &= \nu\left([f - f(p) - \textstyle\sum_{i=1}^n (x_i - x_i(p))h_i(p)]\right) \\ &= \nu([f]) - f(p)\nu([0]) - \textstyle\sum_{i=1}^n \nu([x_i - x_i(p)])h_i(p) \\ &= \nu([f]) - 0 - \textstyle\sum_{i=1}^n c_i\omega_i([f]). \end{aligned}$$

Or in other words, $\nu = \sum_{i=1}^n c_i\omega_i$.

And finally, I claim that $\omega_1,\ldots,\omega_n$ are all linearly independent. After all, suppose $c_1,\ldots,c_n \in \mathbb{R}$ satisfy that $\sum_{i=1}^n c_i\omega_i([f]) = 0$ for every germ $[f]$. Then in particular we know that $c_j = \sum_{i=1}^n c_i\omega_i([x_j - x_j(p)]) = 0$ for every j.

And now to finish the proof, just note for any germ $[(U,f)]$ that:
$$\Phi\left(\left.\frac{\partial}{\partial x_i}\right|_p\right)([f]) = \left.\frac{\partial}{\partial x_i}\right|_p f = \omega_i([f]) \text{ for each } i. \blacksquare$$

For the sake of convenience, I'll identify ω_i from the prior theorem with $\left.\frac{\partial}{\partial x_i}\right|_p$.

One last thing to sort out is how to define the differential of a C^1 map $F : M \to N$ with respect to the vector space $\mathcal{T}_pM$. To do this, we merely say:
$(\mathrm{d}F_p(\nu))([(U,f)]) \coloneqq \nu([(F^{-1}(U), f\circ F)])$ for every $\nu \in \mathcal{T}_pM$ and $[(U,f)] \in C^1_{F(p)}(N)$.

Checking that $\mathrm{d}F_p(\nu)$ is a well-defined operator in $\mathcal{D}_{F(p)}(N)$ for every $\nu \in \mathcal{T}_pM$ and that $\mathrm{d}F_p : \mathcal{T}_pM \to \mathcal{D}_{F(p)}(N)$ is a well-defined linear map is easy.

As for proving that $\mathrm{d}F_p(\nu)$ annihilates all germs vanishing to first order, it suffices to prove that if $[f] \in C^1_{F(p)}(N)$ vanishes to first order then so does $[f \circ F] \in C^1_p(M)$.

To do this, let $[(V, f)]$ be a germ and let $\widetilde{\mathrm{d}}F_p : T_p(F^{-1}(V)) \to T_{F(p)}V$ be the old definition of the differential of F. Then note that $f \circ F$ vanishes to first order iff $v(f \circ F) = 0$ for all $v \in T_p(F^{-1}(V))$. Yet simultaneously $v(f \circ F) = (\mathrm{d}F_p(v))(f)$. And since f vanishes to first order and $\mathrm{d}F_p(v) \in T_{F(p)}V$, we know that:

$$(\mathrm{d}F_p(v))(f) = 0.$$

Also if $\Phi_M : T_pM \to \mathcal{T}_pM$ and $\Phi_N : T_{F(p)}N \to \mathcal{T}_{F(p)}N$ are the maps constructed in the prior theorem, then we can see by looking at a representive for each germ whose domain is all of M or N that:

$$\begin{array}{ccc} & f \mapsto v(f) \qquad\qquad g \mapsto v(g \circ F) & \\ T_pM & \xrightarrow{\widetilde{\mathrm{d}}F_p} & T_{F(p)}N \\ \wr \downarrow \Phi_M & & \wr \downarrow \Phi_N \\ \mathcal{T}_pM & \xrightarrow{\mathrm{d}F_p} & \mathcal{T}_{F(p)}N \\ & [f] \mapsto v(f) \qquad\qquad [g] \mapsto v(g \circ F) & \end{array}$$

Anyways this new definition of the tangent plane is the most general way the tangent plane of a manifold is defined. It is also nice because as mentioned before, it generalizes to types of manifolds (i.e. real-analytic and complex) where germs don't have representatives in any arbitrary neighborhood of a point.

8/17/2026

I'm going to keep being distracted due to the fact I've been in and out of being homeless for two weeks. But right now I kinda want to study again. I'll be studying Lee's book again where I left off.

Local Immersion Theorem For Manifolds with Boundaries: Let M be an m-dimensional C^r manifold with a boundary (but no corners), and let N be an n-dimensional C^r manifold without a boundary. Also suppose $F : M \to N$ is a C^r immersion. For any $p \in \partial M$ there exists a smooth boundary chart (U, φ) on M containing p as well as a smooth chart (V, ψ) on N such that $\varphi(p) = 0$, $F(U) \subseteq V$, and $\widehat{F} := \psi \circ F \circ \varphi^{-1}$ is given by:

$$\widehat{F}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, \underbrace{0, \ldots, 0}_{n-m \text{ zeros}})$$

Proof:
By choosing local coordinates around p it suffices to assume M is an open subset of H^m and N is an open set $\mathbb{R}^n$. This follows by identical reasoning as in the rank theorem. Also by identical reasoning as in the rank theorem, we can assume $p = 0$ and $F(p) = 0$.

Next we know that F extends to a C^r function $\widetilde{F} : W \to \mathbb{R}^n$ such that $M \subseteq W$ and $\widetilde{F}|_M = F$. Because $\mathrm{d}\widetilde{F}_0 = \mathrm{d}F_0$ is an injection (here we recall that we can identify T_pM with T_pW), we can shrink W and M such that $\widetilde{F}$ is a C^r immersion. Also, we can shrink W (without shrinking M) such that $\widetilde{F}(W) \subseteq N$. And now let us write the coordinates on W as $(x_1, \ldots, x_m)$ and the coordinates on N as $((v_1, \ldots, v_m), (w_1, \ldots, w_{n-m}))$.

By the rank theorem (see *page 1062*), there exists smooth charts (U_0, φ_0) and (V_0, ψ_0) such that $0 \in U_0 \subseteq W$, $F(U_0) \subseteq V_0 \subseteq N$, $\varphi_0(0) = 0$, and $\widehat{F} \coloneqq \psi_0 \circ \widetilde{F} \circ \varphi_0^{-1}$ is given by:

$$\widehat{F}(x_1, \ldots, x_m) = (x_1, \ldots, x_m, \underbrace{0, \ldots, 0}_{n-m \text{ zeros}})$$

The one issue that needs to be worked out is that $\varphi_0(U_0 \cap M)$ may not be contained in H^m. To address this, we make the following modification:

φ_0 is a C^r diffeomorphism from $U_0 \subseteq \mathbb{R}^m$ to an open set $\widehat{U}_0 \subseteq \mathbb{R}^m$ such that $\varphi_0(0) = 0$. In turn, $\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}} : \widehat{U}_0 \times \mathbb{R}^{n-m} \to U_0 \times \mathbb{R}^{n-m}$ is a C^r diffeomorphism defined on an open neighborhood of 0.

Next let $\psi \coloneqq (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}}) \circ \psi_0$. This is well-defined C^r diffeomorphism from an open set $V \coloneqq V_0 \cap \psi_0^{-1}(\widehat{U}_0 \times \mathbb{R}_{n-m})$ of N containing $F(0) = 0$.

Finally $U \coloneqq U_0 \cap F^{-1}(V) \cap M$ is an open subset of M containing 0 such that for all $x \in U$:

$$\begin{aligned}\psi \circ F(x_1, \ldots, x_m) &= (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}}) \circ \psi_0 \circ F \circ \varphi_0^{-1} \circ \varphi_0(x_1, \ldots, x_m) \\ &= (\varphi_0^{-1} \times \mathrm{Id}_{\mathbb{R}^{n-m}})(\varphi(x_1, \ldots, x_m), 0) = (x_1, \ldots, x_m, 0, \ldots, 0)\end{aligned}$$

Therefore the original coordinates for M restricted to U as well as the smooth chart (V, ψ) works. ■

Given C^r manifolds M and N, we say a map $F : M \to N$ is a C^k embedding (where $k \leq r$) if F is a C^k immersion and F is a homeomorphism from M to $F(M)$.

Exercise 4.16: If M, N, P are C^r manifolds, $k \leq r$, and $F : M \to N$ and $G : N \to P$ are C^k embeddings then $(G \circ F) : M \to P$ is a C^k embedding as well.

Proof:
$G \circ F$ is obviously C^k. Also a composition of immersions will be an immersion. After all, for all $p \in M$ we have that $\mathrm{d}(G \circ F)_p = \mathrm{d}G_{F(p)} \circ \mathrm{d}F_p$ is injective since $\mathrm{d}F_p$ and $\mathrm{d}G_{F(p)}$ are both injective. Finally, let $h_F : F(M) \to M$ and $h_G : G(N) \to N$ be the continuous inverse functions to F and G respectively. Then $(h_F \circ h_G)|_{G(F(M))}$ is a continuous inverse function to $G \circ F$. ■

Some examples of C^r embeddings are:

1. If M is a C^r manifold and $U \subseteq M$ is an open set, then the inclusion map $\iota : U \hookrightarrow M$ is a C^r embedding.

   It's C^r, at every point it's differential is just the identity linear map, and also it's injective with an inverse map from $\iota(U) \to U$ just being the identity function (which is obviously continuous).

2. Suppose $M_1, \ldots, M_k$ are C^r manifolds and $p_i \in M_i$ is an arbitrarily chosen point for each i. Then for every $1 \leq j \leq k$ we have that the map $\iota_j : M_j \to M_1 \times \cdots \times M_k$ given by $\iota_j(m_j) = (p_1, \ldots, p_{j-1}, m_j, p_{j+1}, \ldots, p_k)$ is a C^r embedding.

   That ι_j is C^r is immediate. Also we know that it has a continuous C^r inverse $\pi_j : M_1 \times \cdots \times M_k \to M_j$ given by $(m_1, \ldots, m_{j-1}, m_j, m_{j+1}, \ldots, m_k) \mapsto m_j$. Finally, for any $q \in M_j$ we have that$\mathrm{d}(\mathrm{Id}_{M_j})_q = \mathrm{d}\pi_{\iota_j(q)} \circ \mathrm{d}(\iota_j)_q$ implies that $\mathrm{d}\pi_{\iota_j(q)}$ is surjective and $\mathrm{d}(\iota_j)_q$ is injective. From this we can conclude that ι_j is an immersion and π_j is a submersion.

Next, here are two nonexamples of smooth embeddings:

1. Consider the map $\gamma : \mathbb{R} \to \mathbb{R}^2$ given by $\gamma(t) = (t^3, 0)$. This is a C^∞ map with a continuous inverse function given by $(t, 0) \mapsto (\sqrt[3]{t}, 0)$. That said, $\mathrm{d}\gamma_0$ is a trivial linear map since $\gamma'(0) = 0$. So, γ fails to be an immersion.

2. Consider the lemniscate $\beta : (-\pi, \pi) \to \mathbb{R}^2$ defined by $\beta(t) = (\sin(2t), \sin(t))$.

   Proving that β is injective just involves considering a bunch of subcases. So suppose $s, t \in (-\pi, \pi)$ satisfy that $\sin(2t) = \sin(2s)$ and $\sin(t) = \sin(s)$.

   - If $t \in (0, \pi/2)$ and $s \neq t$ then $\sin(t) = \sin(s) \implies s = \pi - t$. Yet then $\sin(2s) < 0$ while $\sin(2t) > 0$. So if $t \in (0, \pi/2)$ then $s = t$.
   - If $t \in (\pi/2, \pi)$ and $s \neq t$ then $\sin(t) = \sin(s) \implies s = \pi - t$ again. Yet now we have that $\sin(2t) < 0$ while $\sin(2s) > 0$. So if $t \in (\pi/2, \pi)$ then $s = t$.
   - Given that $t \in (-\pi, \pi)$, we have that $\sin(t) = 0$ iff $t = 0$ and $\sin(t) = 1$ iff $t = \pi/2$. So, β is injective on $[0, \pi)$.
   - By the oddness of $\sin(2t)$ and $\sin(t)$ we can also conclude that β is injective on $(-\pi, 0]$. And since $\sin(t) < 0$ for all $t < 0$ and $\sin(t) > 0$ for all $t > 0$, we can conclude that β is injective on $(-\pi, \pi)$.

   We can also see that β is a C^∞ map. In particular, $\beta'(t) = (2\cos(2t), \cos(t))$. But now $\cos(t) = 0$ iff $t = \pm\pi/2$. Yet $\cos(2(\pi/2)) = -1 = \cos(2(-\pi/2))$. So $\beta'(t) \neq 0$ for any $t \in (-\pi, \pi)$ and we can conclude that β is an immersion.

   That all said, β does not have a continuous inverse function. After all, $\beta(0) = 0$ but also $\beta(t) \to 0$ as $t \to \pi$ or as $t \to -\pi$. Therefore, if $f : \beta((-\pi, \pi)) \to (-\pi, \pi)$ was the inverse function to β then we'd have to have that f is not continuous at 0.

Proposition 4.22: Suppose M and N are C^r manifolds with or without boundary, and $F : M \to N$ is an injective C^k immersion (where $k \leq r$). Then F is a C^k embedding if any of the following conditions hold:

(a) F is an open or closed map into N.

Why?
F being an open map into N trivially means F is an open map into the subspace topology of $F(M) \subseteq N$.

Meanwhile, if F is a closed map then let $U \subseteq M$ be an open set. Then $M - U$ is a closed set so $F(M - U)$ is a closed subset of N. It follows that $N - F(M - U)$ is an open subset of N. And so $F(M) \cap (N - F(M - U)) = F(U)$ (which holds because F is injective) is an open set in the subspace topology of $F(M) \subseteq N$.

More generally, this reasoning shows that if X and Y are topological spaces then any injective continuous map $f : X \to Y$ that is also open or closed is a topological embedding (i.e. a homeomorphism from X to $f(X) \subseteq Y$).

(b) F is a proper map.

Why?
Let E be any closed subset of M and then suppose q is any accumulation point of $F(E)$ in N. Since N is an LCH space, we can pick a precompact open set $V \subseteq N$ containing q. And now $\overline{V} \cap F(E)$ also has q as an accumulation point. Yet because F is proper, we know that $F^{-1}(\overline{V})$ is compact. Then since closed subsets of compact sets are also compact, we know that $E \cap F^{-1}(\overline{V})$ is compact. In turn, as F is continuous we have that $F(E \cap F^{-1}(\overline{V})) = F(E) \cap \overline{V}$ is compact in N (and thus closed since N is Hausdorff). So $F(E) \cap \overline{V}$ must contain all its accumulation points and so $q \in F(E)$.

Since q was arbitrary, we've proven that $F(E)$ is a closed subset of N. So, F is a closed map into N and we can finish by citing part (a).

More generally, this reasoning shows that if X is a topological space and Y is an LCH space then any continuous proper map $f : X \to Y$ is also a closed map.

(c) M is a compact manifold.

Why?
$F : M \to F(N)$ is a continuous bijection, M is compact, and $F(N)$ is Hausdorff. Thus F is a homeomorphism.

(d) M has no boundary and $\dim(M) = \dim(N)$.

Why?
By proposition 4.8(b) on *page 1061*, we know F is a local C^k-diffeomorphism. It then follows by proposition 4.6(c) on the page before proposition 4.8(b) that F is an open map. So, we finish by invoking part (a) of this theorem. ■

8/20/2026

I was realizing yesterday that I've conflated open maps with closed maps an unhealthy amount of times in my brain and am not sure if that's made some of my prior notes incorrect. I've double checked a bunch of pages and not found any errors but still.

Anyways a thing I want to get across first and foremost today is that a closed surjective continuous map is a quotient map for different reasons then an open surjective continuous map is.

Proposition: Suppose X and Y are topological spaces and $f : X \to Y$ is a continuous surjective map. Then f is a quotient map if either of the following holds:

(a) f is an open map.

Why?
Suppose $U \subseteq Y$ satisfies that $f^{-1}(U)$ is an open subset of X. Then because f is surjective, we know that $f(f^{-1}(U)) = U$. And because f is open, we know that U is open.

(b) f is a closed map.

Why?
Suppose $U \subseteq Y$ again satisfies that $f^{-1}(U)$ is an open subset of X. Then we know that $X - f^{-1}(U)$ is a closed subset of X. So because f is a closed map, we know that $f(X - f^{-1}(U)) = f(X) - U$ is a closed subset of Y. Since f is surjective, we in other words know that $Y - U$ is a closed set. Equivalently, U is an open set. ■

Another thing I want to mention is that I should think of a C^k embedding as the differential-topology equivalent of a topological embedding. Similarly, a (surjective) C^k submersion is the differential-topology equivalent of a quotient map as I will soon show.

One final observation I want to make before getting back into taking notes on Lee's book is that we can precisely restate the definition of a coordinate patch α for a k-dimensional C^r manifold in $\mathbb{R}^n$ given on *page 84* as being that α is a C^r-embedding. I will come back to this soon when talking about embedded submanifolds.

Theorem 4.25 (Local Embedding Theorem): Suppose M and N are C^r manifolds where M has no corners and $F : M \to N$ is a C^k map. Then F is a C^k immersion if and only if for all $p \in M$ there exists an open set $U \subseteq M$ such that $F|_U : U \to N$ is a C^k embedding.

$(\Longleftarrow)$
This direction is immediate.

$(\Longrightarrow)$
Suppose F is a C^k immersion. Without loss of generality we can assume M and N are equipped with the unique C^k structures containing their original C^r structures since that would have no effect on whether F is a C^k immersion or $F|_U$ is a C^k embedding. And this assumption is useful cause it let's us apply the rank theorem.

Firstly we show that any $p \in M$ has a neighborhood U on which F is injective.

First suppose $F(p) \notin \partial N$. Then as the set of interior points $\text{Int}(N)$ is an open set and F is continuous, we know that there is an open set $\Omega \subseteq M$ containing p with $F(\Omega) \subseteq \text{Int}(N)$. Then by either the rank theorem on *page 1062* or the local immersion theorem on *page 1101*, we know there is an open set $U_1 \subseteq M$ of p where F has the coordinate representation:

$$(x_1, \ldots, x_m) \mapsto (x_1, \ldots, x_m, 0, \ldots, 0).$$

So F is injective on U_1.

Meanwhile, suppose $F(p) \in \partial N$ and let (W, ψ) be a smooth chart on N containing $F(p)$. Then $U_0 := F^{-1}(W)$ is an open subset of M containing p. Furthermore, let $\iota : H^n (\text{ or } \mathbb{R}^n_{\geq 0}) \hookrightarrow \mathbb{R}^n$ be the inclusion map (here $n = \dim(N)$). Then $\iota \circ \psi \circ F|_{U_0}$ is a C^k map from M to a C^k manifold without boundary. Also, as $\mathrm{d}\iota_a$ and $\mathrm{d}\psi_b$ are bijections for all $a \in H^n$ and $b \in W$, we can conclude that $\iota \circ \psi \circ F|_{U_0}$ is a C^k-immersion still. Thus, we can apply the earlier reasoning to this map to find an open set $U_1 \subseteq U_0$ containing p such that $(\iota \circ \psi \circ F|_{U_0})|_{U_1}$ is an injection. Equivalently, this implies $F|_{U_1}$ is an injection.

Now consider any $p \in M$ and let $U_1 \subseteq M$ be an open set containing p on which F is injective. Since M is an LCH space, we can find a precompact open set U such that $p \in U \subseteq \overline{U} \subseteq U_1$. But now $F|_{\overline{U}}$ is an injective continuous function from a compact set. Therefore, $F|_{\overline{U}}$ is a topological embedding. Next since restrictions of topological embeddings are still topological embeddings, we know $F|_U$ is a topological embedding. And since $F|_U$ is a C^k immersion, we can conclude that $F|_U : U \to N$ is a C^k embedding. ■

Suppose M and N are C^r manifolds. If $\pi : M \to N$ is a continuous map then a section for π is a continuous right inverse function $\sigma : N \to M$ (i.e. $\pi \circ \sigma = \text{Id}_N$). A local section for π is a continuous map $\sigma : U \to M$ where $U \subseteq N$ is an open set and $\pi \circ \sigma = \text{Id}_U$.

Theorem 4.26 (Local Section Theorem): Suppose M and N are C^r manifolds without boundaries, and $\pi : M \to N$ is a C^k map (where $k \leq r$). Then π is a C^k submersion if and only if every $p \in M$ is in the image of a C^k local section of π.

$(\Longleftarrow)$

Given any $p \in M$, let $\sigma : U \to M$ be a C^k local section such that $q \in U$ and $\sigma(q) = p$. In other words, $q = \pi(\sigma(q)) = \pi(p)$. But now since $\pi \circ \sigma = \text{Id}_U$, we can conclude that $\mathrm{d}\pi_p \circ \mathrm{d}\sigma_q = \text{Id}_{T_q N}$. In turn, we must have that $\mathrm{d}\pi_p$ is surjective. And since p was arbitrary, we can conclude π is a submersion.

$(\Longrightarrow)$

Suppose π is a C^k submersion. Then without loss of generality we can assume M and N are equipped with the unique C^k structures containing their original C^r structures. After all, this change won't effect whether π is a C^k submersion, or whether some local section of π is C^k or not. Also, this assumption lets us apply the rank theorem.

Pick any $p \in M$ and let $q = \pi(p)$. By the rank theorem, we can choose smooth charts (V_0, φ) on M containing p and (U_0, ψ) containing q such that $\varphi(p) = 0$, $\pi(V_0) \subseteq U_0$, and $\widehat{\pi} := \psi \circ \pi \circ \varphi^{-1}$ has the form $\widehat{\pi}(x_1, \ldots, x_n, x_{n+1}, \ldots, x_m) = (x_1, \ldots, x_n)$.

For a small enough $\varepsilon > 0$ we have that the coordinate cube
$$C_\varepsilon := \{(x_1, \ldots, x_m) : |x_i| < \varepsilon \text{ for all } i\}$$
is an open subset of $\varphi(V_0)$ containing $0 = \varphi(p)$. In turn we have that:
$$C'_\varepsilon := \{(y_1, \ldots, y_n) : |y_i| < \varepsilon \text{ for all } i\}$$
is an open subset of $\psi(U_0)$ containing $0 = \widehat{\pi}(0) = \widehat{\pi} \circ \varphi(p) = \psi(q)$ and satisfying that $\widehat{\pi}(C_\varepsilon) = C'_\varepsilon$. Let $U = \psi^{-1}(C'_\varepsilon)$. Then define $\iota : C'_\varepsilon \to C_\varepsilon$ by:
$$\iota(y_1, \ldots, y_n) = (y_1, \ldots, y_n, \underbrace{0, \ldots, 0}_{m-n \text{ zeros}}).$$

Finally, let $\sigma = \varphi^{-1} \circ \iota \circ \psi|_U$. Then if $q' \in U$ and $\varphi(q') = (y_1, \ldots, y_n)$ we have that:
$$\begin{aligned} \pi \circ \sigma(q') &= \psi^{-1} \circ \widehat{\pi} \circ \varphi \circ (\varphi^{-1} \circ \iota \circ \psi)(q') \\ &= \psi^{-1} \circ \widehat{\pi} \circ \iota(y_1, \ldots, y_n) = \psi^{-1} \circ \widehat{\pi}(y_1, \ldots, y_n, 0, \ldots, 0) \\ &= \psi^{-1}(y_1, \ldots, y_n) = q' \end{aligned}$$

So $\pi \circ \sigma = \mathrm{Id}_U$. Also σ is a C^k map and
$$\sigma(q) = \varphi^{-1} \circ \iota(0, \ldots, 0) = \varphi^{-1}(0, \ldots, 0) = p. \blacksquare$$

Based on theorems 4.25 and 4.26, we also get the following purely topological definitions. Suppose X and Y are topological spaces and $f : X \to Y$ is a continuous function.

- f is a topological immersion if every point $x \in X$ has a neighborhood $U \subseteq X$ such that $f|_U : U \to Y$ is a topological embedding.
- f is a topological submersion if every point $x \in X$ is in the image of a (continuous) local section for f.

The prior two theorems say that any C^k immersion / submersion is a topological immersion / submersion. That said, be aware that a topological immersion or submersion that happens to also be a C^k map is not necessarily a "C^k immersion" or "C^k submersion".

For example, let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(t) = t^3$. Then f is a C^∞ map which is also a homeomorphism. And clearly homeomorphisms are trivially topological immersions and submersions. That said, as $\mathrm{d}f_0$ is the trivial linear map as $f'(0) = 0$, we know f is not a C^∞ immersion or submersion.

Theorem 4.27: Let M and N be C^r manifolds without boundaries, and suppose $\pi : M \to N$ is a C^k submersion. Then π is an open map. Also consequently, if π is surjective then π is a quotient map.

Proof:
Suppose $W \subseteq M$ is an open set and $q \in \pi(W) \subseteq N$. We can pick $p \in W$ with $\pi(p) = q$. And then by the local section theorem, we know there is an open subset $U \subseteq N$ containing q as well as a C^k map $\sigma : U \to M$ such that $\sigma(q) = p$ and $\pi \circ \sigma = \mathrm{Id}_U$. Yet now note

that $\sigma^{-1}(W)$ is an open subset of U (since σ is continuous) and it contains q. Also note that if $y \in \sigma^{-1}(W)$ then $y = \pi(\sigma(y)) \in \pi(W)$ Therefore, we've shown that q is in the topological interior of $\pi(W)$. And since q was arbitrary, this proves that $\pi(W)$ is an open set. ■

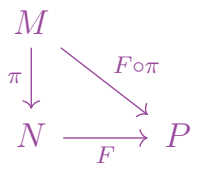
Exercise 4-6: Let M be a nonempty compact C^r manifold without a boundary, and let N be a connected non-compact C^r without a boundary. Then there is no C^k submersion $\pi : M \to N$ (where $k \leq r$).

Proof:
By the prior theorem we must have that such a function π is an open map. Thus in particular we can conclude that $\pi(M)$ is an open subset of N. That said, as M is compact and π is continuous, we must also have that $\pi(M)$ is a compact subset of N. Since N is Hausdorff, we can conclude $\pi(M)$ is a clopen set. And since N is connected, we thus must have that $\pi(M)$ is either the empty set or all of N.

$\pi(M)$ can't be the empty set however because $M \neq \emptyset$. Simultaneously, since N is not compact we can't have that $\pi(M) = N$. Thus we have a contradiction. ■

The next theorem is a generalization of exercise 4.10(b) on *pages 1061-1062*.

Theorem 4.29: Suppose M and N are C^r manifolds without boundaries and $\pi : M \to N$ is a surjective C^k submersion. For any C^r manifold P we have that any map $F : N \to P$ is C^k iff $F \circ \pi : M \to P$ is a C^k map.



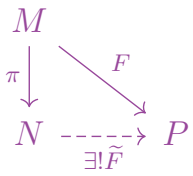
Proof:
If F is C^k then the composition $F \circ \pi$ will also be C^k. So, one direction of this theorem is trivial.

Conversely, suppose $F \circ \pi$ is C^k and let $q \in N$ be any point. Then as π is surjective we know that there exists $p \in M$ with $\pi(p) = q$. By the local section theorem, we know there exists an open set $U \subseteq N$ containing q as well as a C^k map $\sigma : U \to M$ such that $\sigma(q) = p$ and $\pi \circ \sigma = \mathrm{Id}_U$. And now $F|_U = F \circ \mathrm{Id}_U = F \circ (\pi \circ \sigma) = (F \circ \pi) \circ \sigma$ is a C^k map.

Since every point in N has a neighborhood on which F is a C^k map, we can conclude that F is globally a C^k map. ■

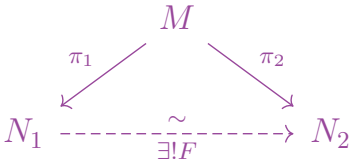
Corollary 4.30: Suppose M and N are C^r manifolds without boundaries and $\pi : M \to N$ is a surjective C^k submersion. If P is any C^r manifold and $F : M \to P$ is any C^k map that is constant on the fibers of π (meaning F is constant on each set $\pi^{-1}(q)$ for every $q \in N$),

then there exists a unique C^k map $\widetilde{F} : N \to P$ such that $F = \widetilde{F} \circ \pi$.

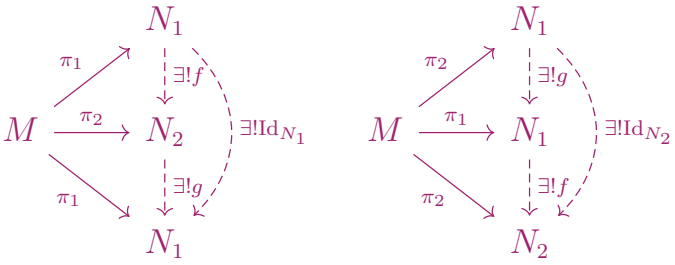


Proof:
Since π is a quotient map, we know from the universal property of quotient maps that there exists a unique continuous function $\widetilde{F} : N \to P$ such that $F = \widetilde{F} \circ \pi$. Then as F is C^k, the prior theorem tells us that $\widetilde{F}$ is also C^k. ■

Corollary 4.31: Suppose M, N_1, N_2 are C^r manifolds without boundaries and $\pi_1 : M \to N_1$ and $\pi_2 : M \to N_2$ are surjective C^k submersions that are constant on each other's fibers (i.e. $\pi_1(x) = \pi_1(y) \iff \pi_2(x) = \pi_2(y)$). Then there exists a unique C^k diffeomorphism $F : N_1 \to N_2$ such that $\pi_2 = F \circ \pi_1$.



Why?
Just consider the following two diagrams:



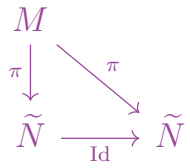
Something I want to quickly mention is that an anlogous corollary can be proven for quotient maps $p_1 : X \to Y_1$ and $p_2 : X \to Y_2$ (where X, Y_1, Y_2 are topological spaces). My original learning about quotient maps was fragmented so I never wrote that down before, but I think it's worth remembering in the future.

One last exercise I want to do before forcing myself to try to sleep is as follows:

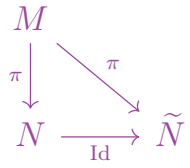
Exercise 4-7: Let M and N be C^r manifolds without boundaries, and let $\pi : M \to N$ be a surjective C^r submersion. Show there is no other manifold topology and C^r structure that we can equip N with such that for any C^r manifold P and map $F : N \to P$ we have that F is C^r if and only if $F \circ \pi$ is C^r.

Proof:
Let $\widetilde{N}$ be the same set as N except equipped with a potentially different manifold topology and C^r structure. Since the following diagram commutes and Id is a C^r map from $\widetilde{N}$ to $\widetilde{N}$,

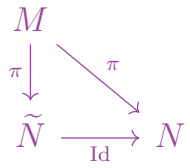
we can conclude that $\pi : M \to \widetilde{N}$ is a C^r map.



But then since $\pi : M \to \widetilde{N}$ is a C^r map we can also conclude from the following diagram that $\mathrm{Id} : N \to \widetilde{N}$ is a C^r map.



Meanwhile, as $\pi : M \to N$ is a C^r map we can conclude from the following diagram that $\mathrm{Id} : \widetilde{N} \to N$ is a C^r map.



So, we've proven that $\mathrm{Id} : N \to \widetilde{N}$ is a C^r-diffeomorphism. Since this also means Id is a homeomorphism, we can conclude that the topologies on N and $\widetilde{N}$ are identical. Also, $\mathrm{Id} : N \to \widetilde{N}$ being a C^r diffeomorphism easily lets us conclude that every smooth chart on N is C^r compatible with every smooth chart on $\widetilde{N}$ and vice versa.

Why?
Suppose (U, φ) is a smooth chart with respect to N and (V, ψ) is a smooth chart with respect to $\widetilde{N}$ such that $U \cap V \neq \emptyset$. Then as $\mathrm{Id} : N \to \widetilde{N}$ is a C^r-diffeomorphism, we know that $\psi \circ \varphi^{-1} = \psi \circ \mathrm{Id} \circ \varphi^{-1}$ and $\varphi \circ \psi^{-1} = \varphi \circ \mathrm{Id} \circ \psi^{-1}$ are C^r maps.

It follows that the C^r structures on N and $\widetilde{N}$ are also the same. ■

In particular, this shows that if M is a C^r manifold without a boundary and $\pi : M \to N$ is any surjective function then there is at most one manifold topology without boundary and C^r structure on N such that π is a C^r submersion.

Chapter 5 of John Lee's book will let us fully connect the abstract definition of a manifold with the more naive definition studied on *pages 77-99*.

Suppose M is a C^r manifold. A C^r embedded submanifold of M is a subset $S \subseteq M$ which is a topological manifold when equipped with the subspace topology and which has a C^r structure such that the inclusion map $\iota : S \to M$ is a C^r embedding.

Remarks:

- If $k \leq r$ then M has a unique C^k structure containing its C^r structure. Based on this structure on M, if S is a C^k manifold then we can ask if $\iota : S \to M$ is a C^k embedding. And if so we call S a C^k embedded submanifold of M.
- Some authors also refer to embedded submanifolds as regular submanifolds.

If M is a C^r manifold and $S \subseteq M$ is a C^k embedded submanifold of M then the codimension of S in M is defined to be the difference $\dim(M) - \dim(S)$. Here M is called the ambient manifold for S. If S has codimension 1 we say S is an embedded hypersurface.

Proposition 5.1: Suppose M is a C^r manifold without a boundary. Then the embedded C^k submanifolds of codimension 0 (without boundaries) in M are precisely the open submanifolds of M.

Proof:
If $U \subseteq M$ is an open submanfold then we know $\dim(U) = \dim(M)$. Furthermore, U is equipped with the subspace topology and we know from before that the inclusion map $\iota : U \hookrightarrow M$ is a C^k immersion and submersion. It follows that ι is a C^k embedding. And as M has no boundary, neither does U.

Conversely, suppose $U \subseteq M$ is an embedded C^k submanifold in M of codimension 0. By definition the inclusion map $\iota : U \hookrightarrow M$ is a C^k embedding. Yet as $\dim(M) = \dim(U)$ and ι is an immersion, we know from proposition 4.8(b) on *page 1061* that ι is a local C^k-diffeomorphism. In turn, by proposition 4.6(c) on *page 1060* we know ι is an open map. So, $\iota(U) = U$ is an open subset of M.

Finally, suppose (V, φ) is a smooth chart on M (equipped with its unique C^k structure). Then $U \cap V$ is an open subset of U and I claim that $(U \cap V, \varphi)$ is a smooth map on U. After all, suppose (W, ψ) is any other smooth chart on U as an embedded submanifold. Then as $\iota : U \hookrightarrow M$ is a C^k map we know that $\varphi \circ \iota \circ \psi^{-1} = \varphi \circ \psi^{-1}$ is C^k on $\psi((U \cap V) \cap W)$.

Meanwhile, as ι is a local C^k-diffeomorphism, we know for every $p \in (U \cap V) \cap W$ that there is some open set $\Omega \subseteq U$ containing p such that ι is a C^k-diffeomorphism from Ω (as an open submanifold of U) to Ω (as an open submanifold of M). In turn, $\psi \circ \varphi^{-1}$ is a C^k map on $\psi((U \cap V) \cap W \cap \Omega)$. So, $\psi \circ \varphi^{-1}$ is C^k at every $p \in (U \cap V) \cap W$. And with that, we've proven that the C^k structure on U as an embedded submanifold must be the same as the C^k structure on U as an open submanifold. ■

Proposition 5.2: Suppose M and N are C^r manifolds, and $F : N \to M$ is a C^r embedding. Let $S = F(N)$. With the subspace topology S is a topological manifold. Also S has a unique C^r structure making it into an embedded submanifold of M and satisfying that F is a C^r-diffeomorphism from N to S.

Proof:
S being a subspace of M and M being second countable and Hausdorff means that S is also second countable and Hausdorff. Also by assumption F is a homeomorphism from N to S. Therefore, N being locally Euclidean of dimension $\dim(N)$ means that S is as well.

Now we equip S with the C^r structure given by the charts $(F(U), \varphi \circ F^{-1})$ where (U, φ) is any smooth chart on N. Them being C^r compatible with each other is a result of the smooth charts on N being C^r compatible with each other. So, S is a C^r manifold. It's also easy to see now that F is a C^r-diffeomorphism. After all, for any $p \in S$ we can pick a smooth chart (U, φ) on N containing $F^{-1}(p)$. Then $(F(U), \varphi \circ F^{-1})$ is a smooth chart on S containing p, and:

$$\varphi \circ F \circ (\varphi \circ F^{-1})^{-1} = \mathrm{Id}_{\varphi(U)} \text{ is a } C^r \text{ map.}$$

So, $F^{-1} : S \to N$ is a C^r map.

Next, to show uniqueness just note that if F is to be a diffeomorphism from N to S then for any smooth chart (U, φ) on N we must have that $(F(U), \varphi \circ F^{-1})$ is a smooth chart on S. Similarly, for every smooth chart (V, ψ) on S we must have that $(F^{-1}(V), \psi \circ F)$ is a smooth chart on N. So the atlas described in the prior paragraph is maximal and F being a C^r-diffeomorphism from N to S automatically implies that S is equipped with that maximal structure.

Finally, I need to show that the map $\iota : S \to M$ is a C^r-embedding. Fortunately, consider that ι is given by the composition:

$$S \xrightarrow{F^{-1}} N \xrightarrow{F} M \qquad (\iota : S \to M)$$

Since F^{-1} is a C^r-diffeomorphism and F is a C^r-embedding, we can conclude ι is a C^r embedding. So, S is an embedded submanifold of M. ■

> As a side note, if $F : N \to M$ is a C^k embedding (where $k \leq r$) we can also now say that $S = F(N)$ has a unique C^k structure making it into an embedded submanifold of M such that F is a C^k-diffeomorphism from N to S. After all, we can just re-equip M and N with the unique C^k structures containing their original C^r structures and then everything works the same.

Proposition 5.3: Suppose M and N are C^r manifolds. For each $p \in N$ the subset $M \times \{p\} \subseteq M \times N$ (called a slice of the product manifold) is an embedded submanifold of $M \times N$ diffeomorphic to M.

Proof:
The map $F : M \to M \times N$ given by $F(m) := (m, p)$ is a C^r embedding. After all, it's a C^r map that is a topological embedding. Furthermore, recall from *page 1059* how we standardly identify $T_{(m,p)}(M \times N)$ with $T_m M \oplus T_p N$.

Lemma: Suppose $N, M_1, \ldots, M_k$ are C^r manifolds. Then let $M := \prod_{i=1}^k M_i$ and for each i let $\pi_i : M \to M_i$ be the projection map. Given any $(p_1, \ldots, p_k) \in M$ we identify $T_{(p_1,\ldots,p_k)} M$ with $\bigoplus_{i=1}^k T_{p_i} M_i$ via the standard identification. Now suppose for each $1 \leq i \leq k$ that $F_i : N \to M_i$ is a C^1 map. Then let $F : N \to M$ be defined by $F(q) = (F_1(q), \ldots, F_k(q))$. Now for any $q \in N$ we have that:

$$\mathrm{d}F_q(v) = (\mathrm{d}(F_1)_q(v), \ \ldots, \ \mathrm{d}(F_k)_q(v)).$$

Why?
Suppose $v \in T_qN$ and $f : M \to \mathbb{R}$ is a C^1 map. Then $\mathrm{d}F_q(v)f$ is identified with:

$$(\mathrm{d}(\pi_1)_{F(q)} \circ \mathrm{d}F_q(v), \ldots \mathrm{d}(\pi_k)_{F(q)} \circ \mathrm{d}F_q(v)) = (\mathrm{d}(\pi_1 \circ F)_q(v), \ldots, \mathrm{d}(\pi_k \circ F)_q(v))$$
$$= (\mathrm{d}(F_1)_q(v), \ \ldots, \ \mathrm{d}(F_k)_q(v))$$

Based on the prior lemma, we can see for any $v \in T_mM$ that:

$$\mathrm{d}F_m(v) = (v, 0) \in T_mM \oplus T_pN.$$

It follows that $\mathrm{d}F_m$ is injective for all $m \in M$. So, F is an immersion. ■

Proposition 5.4: Suppose M is a C^r m-dimensional manifold, N is a C^r n-dimensional manifold, $U \subseteq M$ is open, and $f : U \to N$ is a C^k map. Let $\Gamma(f) \subseteq M \times N$ denote the graph of f:

$$\Gamma(f) \coloneqq \{(m, f(M)) \in M \times N : m \in U\}$$

Then $\Gamma(f)$ is a C^k m-dimensional embedded submanifold of $M \times N$.

Proof:
Define a map $\gamma_f : U \to M \times N$ by $\gamma_f(m) = (m, f(m))$. This will be a C^k map because the inclusion map $\iota : U \to M$ is C^r and similarly the map $f : U \to N$ is C^k. Also let $\pi_M : M \times N \to M$ be the projection map. Then $\mathrm{Id}_U = \pi_M \circ \gamma_f$ implies that γ_f injective. Also for any $m \in U$ we have $\mathrm{d}(\pi_M)_{(m,f(m))} \circ \mathrm{d}(\gamma_f)_m = \mathrm{d}(\mathrm{Id}_U)_m = \mathrm{Id}_{T_mU}$. So, for every $m \in U$ we have that $\mathrm{d}(\gamma_f)_m$ is injective. In other words, γ_f is an immersion. Finally, γ_f is a topological embedding since its inverse function is given by $\pi_M|_{\Gamma(f)}$.

And now we've shown that $\Gamma(f)$ is the image of a C^k embedding γ_f from U. ■

An embedded submanifold $S \subseteq M$ is said to be properly embedded if the inclusion map $S \hookrightarrow M$ is a proper map.

Proposition 5.5: Suppose M is a C^r manifold and $S \subseteq M$ is an embedded submanifold. S is properly embedded if and only if S is a closed subset of M.

Proof:
Suppose the inclusion map $\iota : S \to M$ is a proper map. Then we know that it is a closed map (see *page 1104*). In turn, $S = \iota(S)$ must be a closed subset of M.

Conversely, note that ι being a topological embedding with a closed image implies that ι is proper.

This is because if $K \subseteq M$ is a compact set, then $S \cap K$ will also be compact since S is closed. And now as ι^{-1} is a continuous map, we know that $\iota^{-1}(K) = \iota^{-1}(S \cap K)$ is compact. So, ι is proper. ■

Corollary 5.6: Every compact embedded submanifold is properly embedded.

Proof:
Compact subsets of Haudorff spaces are closed. ■

Proposition 5.7: Suppose M and N are C^r manifolds, and $f : M \to N$ is a C^k map. Then after equipping $\Gamma(f) \coloneqq \{(m, f(M)) \in M \times N : m \in M\}$ with the smooth structure defined in proposition 5.4, we have that $\Gamma(f)$ is properly embedded.

Proof:
Note that $\gamma_f : M \to M \times N$ given by $\gamma_f(m) = (m, f(m))$ has the map $\pi_M : M \times N \to M$ given by $\pi_M(m, n) = m$ as a continuous left inverse. In other words, $\pi_M \circ \gamma_f = \mathrm{Id}_M$. As a result we can conclude that $\gamma_f : M \to M \times N$ is a proper map. To see why, I claim the following:

Lemma: If X, Y are topological spaces and $g : X \to Y$ and $h : Y \to X$ are continuous functions such that $h \circ g = \mathrm{Id}_X$, then g must be proper.

Why?
Suppose $K \subseteq Y$ is a compact set. Since h is continuous, we know that $h(K) \subseteq X$ is a compact set. But also let $F \coloneqq g^{-1}(K)$. Then:

$$\begin{aligned} F = \mathrm{Id}_X(F) = (h \circ g)(F) &= h(K \cap g(X)) \\ &= h(K) \cap (h \circ g)(X) \\ &= h(K) \cap \mathrm{Id}_X(X) = h(K) \cap X = h(K). \end{aligned}$$

So, we've shown that $g^{-1}(K)$ is compact.

Since γ_f is a continuous proper map though and $M \times N$ is an LCH space, we can conclude that γ_f is a closed map. So $\gamma_f(M) = \Gamma(f)$ is a closed subset of $M \times N$. And by proposition 5.5, we are done. ■

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If $U \subseteq \mathbb{R}^n$ is an open set and $k \in \{0, 1, \ldots, n\}$ then a k-slice of U is a set of the form

$$S \coloneqq \{(x_1, \ldots, x_k, x_{k+1}, \ldots, x_n) \in U : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\}$$

where $c_{k+1}, \ldots, c_n \in \mathbb{R}$. In practice we'll usually have $c_{k+1}, \ldots, c_n = 0$.

More generally, if M is an n-dimensional C^r manifold without a boundary and (U, φ) is a smooth chart on M, then we say a set $S \subseteq U$ is a k-slice of U if $\varphi(S)$ is a k-slice of $\varphi(U) \subseteq \mathbb{R}^n$.

Given a subset S of a C^r n-dimensional manifold M and a nonnegative integer k, we say S satisfies the local k-slice condition if every $p \in S$ is contained in a smooth chart (U, φ) such that $S \cap U$ is a k-slice of U. Such a chart (U, φ) is then called a slice chart and the corresponding local coordinates are called slice coordinates.

Theorem 5.8 (Local Slice Criterion): Let M be a C^r n-dimensional manifold without a boundary.

- If $S \subseteq M$ is an embedded k-dimensional C^r submanifold without a boundary then S satisfies the local k-slice condition.

  Proof:
  Let $p \in S$. Since the inclusion map $\iota : S \hookrightarrow M$ is an immersion, we can use the rank theorem to conclude that there are smooth charts (U, φ) on S and (V, ψ) on M such that $p \in U$, $U \subseteq V \cap S$, $\varphi(p) = 0$, and:

$$\psi \circ \iota \circ \varphi^{-1}(x_1, \ldots, x_k) = (x_1, \ldots, x_k, \underbrace{0, \ldots, 0}_{n-k \text{ zeros}}).$$

Choose $\varepsilon > 0$ small enough so that U and V both contain coordinate cubes $(-\varepsilon, \varepsilon)^k$ and $(-\varepsilon, \varepsilon)^n$ respectively centered at p and denote these coordinate balls as U_0 and V_0. Then U_0 is a k-slice of V_0.

Finally, as S has the subspace topology, we know there is an open set $W \subseteq M$ such that $W \cap S = U_0$. Set $V_1 = V_0 \cap W$. Then $(V_1, \psi|_{V_1})$ is a smooth chart for M containing p such that $V_1 \cap S = U_0$ and U_0 is still a k-slice of V_1.

As a side note if $\pi : \mathbb{R}^n \to \mathbb{R}^k$ is the projection map onto the first k coordinates, then $\varphi = \pi \circ \psi|_{U_0}$. So we've proven for all $p \in S$ can find a slice chart (Ω, ϕ) on M containing p such that $(\Omega \cap S, \pi \circ \phi)$ is a smooth chart on S containing p. This observation will be referenced later on *page* ____.

- Conversely, if $S \subseteq M$ is a subset satisfying the local k-slice condition, then it is a k-dimensional topological manifold without a boundary when equipped with the subspace topology relative to M. Also, there is a C^r structure on S which makes S into a C^r embedded submanifold of M.

  Proof:
  If we give S the subspace topology, then we know S is second countable and Hausdorff because M is. Meanwhile, the idea for constructing an atlas for S is that if $(x_1, \ldots, x_n)$ is the coordinates of a slice chart for S then we can define a chart with the coordinates $(x_1, \ldots, x_k)$.

  We let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates. Now suppose (U, φ) is any slice chart on M. We let:

  $$V \coloneqq U \cap S, \widehat{V} \coloneqq (\pi \circ \varphi)(V), \text{ and } \psi \coloneqq \pi \circ \varphi|_V.$$

  Note that as S is equipped with the subspace topology, V is an open subset of S. Also, by the definition of a slice chart, $\varphi(V) = A \cap \varphi(U)$ for some set:

  $$A = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\}$$

  where $c_{k+1}, \ldots, c_n \in \mathbb{R}$ are constants. It follows that $\varphi(V)$ is an open subset of A. And since $\pi|_A : A \to \mathbb{R}^k$ is a C^∞ diffeomorphism from A to $\mathbb{R}^k$, we know that $\widehat{V} = \pi(\varphi(V))$ is an open subset of $\mathbb{R}^k$.

  Moreover note that $\psi : V \to \widehat{V}$ has a continuous inverse function $\varphi^{-1} \circ j|_{\widehat{V}}$ where $j : \mathbb{R}^k \to \mathbb{R}^n$ is the C^∞ function $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$. So, (V, ψ) is a well-defined interior chart on S.

  Since $S \subseteq M$ has the local k-slice condition, we know every $p \in S$ is contained in some chart constructed above. So, S is a topological manifold without a boundary. Also trivially we have that the inclusion map $\iota : S \hookrightarrow M$ is a topological embedding.

To equip S with a C^r structure, we prove that the charts on S constructed priorly are all C^r compatible with each other. So let (U, φ) and (U', φ') be two different slice charts for S in M and then let (V, ψ) and (V', ψ') be the corresponding charts for S constructed priorly. The transition map $\psi' \circ \psi^{-1}$ will be given by $\pi \circ \varphi \circ \varphi^{-1} \circ j$ where $j : \mathbb{R}^k \to \mathbb{R}^n$ is the map defined priorly. Thus, $\psi' \circ \psi^{-1}$ is a C^r map since it is a composition of four C^r maps. And since ψ and ψ' were arbitrary, we've proven that our priorly constructed atlas defines a C^r structure on S.

Finally, consider any $p \in S$ and let (U, φ) be a slice chart for S in M containing p. Then let (V, ψ) be the corresponding smooth chart on S. We know $p \in V$, $\iota(V) \subseteq U$ and also that there exists $c_{k+1}, \ldots, c_n \in \mathbb{R}$ such that $\varphi \circ \iota \circ \psi^{-1}$ is given by $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$. It follows that ι is a C^r map. And by looking at the Jacobian of $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$, we can also conclude that $\mathrm{d}\iota_p$ is full rank (and injective). ■

As a side note, the local k-slice condition is importantly a local property. One illustration of this fact is as follows:

Suppose M is a C^r n-dimensional manifold and $S \subseteq M$ is a set equipped with the subspace topology on M such that for every $p \in S$ there is an open set $U \subseteq M - \partial M$ containing p such that $S \cap U$ is a k-dimensional embedded C^r submanifold without a boundary. Then S is a k-dimensional topological manifold without a boundary that can be equipped with a C^r structure such that S is an embedded C^r submanifold of M.

This is because $S \cap U$ being an embedded submanifold of M without a boundary implies that we can find a slice chart (V, φ) for S in $U \subseteq M$ containing p. And since p was arbitrary, S satisfies the local k-slice condition.

I'll later show that if M is a C^r manifold without a boundary and $S \subseteq M$ is equipped with the subspace topology then the C^r structure making S into an embedded submanifold (without a boundary) is unique if it exists.

Theorem 5.11: Let M be a C^r n-dimensional manifold without corners. Then ∂M equipped with the subspace topology is an $(n-1)$-dimensional topological manifold without a boundary that can be equipped with a C^r structure such that ∂M is an embedded submanifold of M.

Proof:
∂M is trivially Hausdorff and second countable when equipped with the subspace topology, and similarly the inclusion map $\iota : \partial M \hookrightarrow M$ is trivially a topological embedding when ∂M is equipped with the subspace topology.

To show that ∂M is locally Euclidean, we construct an atlas. Let $\pi : \mathbb{R}^n \to \mathbb{R}^{n-1}$ be given by $\pi(x_1, \ldots, x_n) = (x_1, \ldots, x_{n-1})$ and let $j : \mathbb{R}^{n-1} \to \mathbb{R}^n$ be given by:

$$j(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, 0).$$

Now given any smooth boundary chart (U, φ) for M, we define a chart (V, ψ) on ∂M by $V = U \cap \partial M$ and $\psi = \pi \circ \varphi|_V$.

By the invariance of the boundary, we know that:

$$\varphi(U \cap \partial M) = \{(x_1, \ldots, x_n) \in \varphi(U) : x_n = 0\} = \varphi(U) \cap \partial H^n.$$

It follows that $\varphi(U \cap \partial M)$ is an open subset of ∂H^n. And since π is a homeomorphism from ∂H^n to $\mathbb{R}^{n-1}$, we can conclude that $\psi(V)$ is an open subset of $\mathbb{R}^{n-1}$.

Furthermore, note that ψ has the continuous inverse $\varphi^{-1} \circ j|_{\psi(V)}$. So, ψ is a homeomorphism. And since every $p \in \partial M$ is contained in some boundary chart, we can conclude that the charts constructed here form an atlas for ∂M.

By identical reasoning as in the prior theorem, we can show that the charts constructed prior are C^r compatible with each other, and also that ∂M equipped with the C^r structure they generate is an embedded submanifold of M. ■

There isn't really a way to adapt the k-slice condition to an ambient manifold M with a boundary and so usually the strategy people will instead use to study those manifolds is that they will embed an ambient manifold with a boundary into a larger ambient manifold without a boundary. I'll talk about that way in the future.

Suppose M and N are C^k manifolds without boundaries, $\Phi : M \to N$ is any map, and $c \in N$ is any point. The set $\Phi^{-1}(\{c\})$ is called a level set of Φ. If $N = \mathbb{R}^n$ and $c = 0$, the level set $\Phi^{-1}(\{c\})$ is called the zero set of Φ.

In general, a level set or zero set of a C^k map is not going to be a submanifold.

Theorem 2.29: Let M be a C^k manifold and suppose $K \subseteq M$ is any closed set. Then there is a C^k nonnegative function $f : K \to \mathbb{R}$ such that $f^{-1}(\{0\}) = K$.

Proof:

We start by considering the case where $M = \mathbb{R}^n$ and $K \subseteq \mathbb{R}^n$ is a closed set. Then for every $x \in K^C$ let $r_x \in (0,1)$ be such that $B_{r_x}(x) \subseteq K^C$. Thus $\{B_{r_x/2}(x) : x \in K^C\}$ is an open cover of K^C, and since K^C is second countable, we know there is a countable subcover. So, we've proven there are sequences $\{x_i\}_{i\in\mathbb{N}} \subseteq \mathbb{R}^n$ and $\{r_i\}_{i\in\mathbb{N}} \subseteq (0,1)$ satisfying that:

$$K^C = \bigcup_{i=1}^{\infty} B_{r_i/2}(x_i) = \bigcup_{i=1}^{\infty} B_{r_i}(x_i).$$

Now let $h : \mathbb{R}^n \to [0,1]$ be a C^∞ function such that $h \equiv 1$ on $B_{1/2}(0)$ and $\operatorname{supp}(h) \subseteq B_1(0)$. Also for each $i \in \mathbb{Z}_{\geq 0}$ let $C_i \geq 1$ be some constant such that $\|\partial^\alpha h\|_u \leq C_i$ whenever $|\alpha| \leq i$. And finally, define $f : \mathbb{R}^n \to [0,1]$ by:

$$f(x) = \sum_{i=1}^{\infty} \frac{(r_i)^i}{2^i C_i} h\left(\frac{x - x_i}{r_i}\right)$$

More generally, for any multi-index α let $g_\alpha(x) = \sum_{i=1}^{\infty} \frac{(r_i)^{i-|\alpha|}}{2^i C_i} \partial^\alpha h\left(\frac{x-x_i}{r_i}\right)$. Clearly $g_{(0,\ldots,0)} = f$. By the Weierstrass M-test we can see that each g_α is continuous. Also I claim $\partial^\alpha f = g_\alpha$ for every multi-index α. This specifically follows as an immediate consequence of the following lemma.

Lemma: Suppose $U \subseteq \mathbb{R}^n$ is an open set and let $e_1, \ldots, e_n$ be the standard basis for $\mathbb{R}^n$. Also denote $\partial_j = \frac{\partial}{\partial x_j}$. Next suppose $\{f_k\}_{k\in\mathbb{N}} \in C^1(U, \mathbb{R})$, and also that $g, h \in C(U, \mathbb{R})$ satisfy that $f_k \to g$ pointwise and $\partial_j f_k \to h$ uniformly on compact sets. Then $\partial_j g = h$.

Why?

Let $x \in U$ be any point. Then for small enough $t > 0$ we have that $B_t(x) \subseteq U$. And then by the fundamental theorem of calculus we have that:

$$f_k(x + te_j) - f_k(x) = \int_0^t \partial_j f_k(x + se_j)\mathrm{d}s$$

Since $[-t, t]$ is compact, $\partial_j f$ is continuous, and $\partial_j f_k \to h$ uniformly, we can conclude by the dominated convergence theorem that:

$$\lim_{k\to\infty} \int_0^t \partial_j f_k(x + se_j)\mathrm{d}s = \int_0^t h(x + se_j)\mathrm{d}s.$$

Also since $f_k \to g$ pointwise we know:

$$g(x + te_j) - g(x) = \int_0^t h(x + se_j)\mathrm{d}s$$

And now by the fundamental theorem of calculus, we can conclude that $\frac{\mathrm{d}}{\mathrm{d}t}(g(x + te_j)) = h(x + te_j)$ exists and equals h when $|t|$ is small enough. In other words, $\partial_j g = h$ at x.

By the prior lemma, we can conclude that if β and α are multi-indices such that $\partial^\alpha = \frac{\partial}{\partial x_j} \circ \partial^\beta$ then $\frac{\partial}{\partial x_j} g_\beta = g_\alpha$. Then the fact that $\partial^\alpha f = g_\alpha$ immediately follows by induction.

To finish, off, note that if $x \in K$ then $x \notin B_{r_i}(x_i)$ for any i and so $f(x_i) = 0$. Meanwhile, if $x \notin K$ then $x \in B_{r_i/2}(x_i)$ for some i and hence $h(\frac{x - x_i}{r_i}) = 1$. It follows that $f(x) > 0$. So, $K = f^{-1}(\{0\})$.

Now that this theorem has been shown in the case that $M = \mathbb{R}^n$, we can now consider the general case for a C^k manifold. Suppose $K \subseteq M$ is any closed set and let $\{(B_\alpha, \varphi_\alpha)\}_{\alpha\in A}$ be a covering of M with smooth coordinate balls, half balls, and corner balls. We can let $\{\psi_\alpha\}_{\alpha\in A}$ be a C^r subordinate partition of unity as constructed on *page 1032*.

Note that $B_\alpha \cap K$ is a closed subset relative to B_α for every $\alpha \in A$. In turn, we can see that $\varphi(B_\alpha \cap K)$ is closed relative to $\varphi(B_\alpha)$. Then there is some closed set $F_\alpha \subseteq \mathbb{R}^n$ such that $F_\alpha \cap \varphi(B_\alpha) = \varphi(B_\alpha \cap K)$. And now by the prior reasoning, we let $f_\alpha : \mathbb{R}^n \to [0, \infty)$ be a C^∞ map such that $f_\alpha^{-1}(\{0\}) = F_\alpha$.

But now $f = \sum_{\alpha\in A} \psi_\alpha \cdot (f_\alpha \circ \varphi_\alpha)$ defines a C^r map on M with $f^{-1}(\{0\}) = K$. ■

However, by making assumptions about the rank of a C^k map Φ, we can guarentee that the level sets of Φ are embedded submanifolds.

Theorem 5.12 (Constant-Rank Level Set Theorem): Let M and N be C^k manifolds without boundaries and let $\Phi : M \to N$ be a C^k map with constant rank r. Then each level set of Φ is a properly embedded submanifold of codimension r in M without a boundary.

Proof:

Write $m = \dim(M)$ and $n = \dim(N)$. Also let $c \in N$ be arbitrary and let $S \coloneqq \Phi^{-1}(\{c\}) \subseteq M$. Now for any $p \in S$, the rank theorem gives us smooth charts (U, φ) on M and (V, ψ) on N such that $\varphi(p) = 0$ and:

$$\psi \circ \Phi \circ \varphi^{-1}(x_1, \ldots, x_r, x_{r+1}, \ldots, x_m) = (x_1, \ldots, x_r, \underbrace{0, \ldots, 0}_{n-r \text{ zeros}}).$$

Yet note that $\Psi(p) = c$ implies that $\psi(c) = 0$. And now:

$$U \cap S = U \cap \Phi^{-1}(\{c\}) = \varphi^{-1}(\{(x_1, \ldots, x_m) \in \varphi(U) : x_1 = \cdots = x_r = 0\}).$$

It follows that $U \cap S$ is a $(m - r)$-slice of (U, φ). Since $p \in S$ was arbitrary, we can conclude that S satisfies the local $(m-r)$-slice condition. So, S is an embedded $(m-r)$-dimensional C^k submanifold of M without a boundary.

Also as S is a closed set (since $\{c\}$ is a closed subset of N and Φ is continuous), we know S is properly embedded. ■

Corollary 5.13 (Submersion Level Set Theorem): If M and N are C^k manifolds without boundaries and $\Phi : M \to N$ is a C^k submersion, then each level set of Φ is a properly embedded submanifold of M without a boundary whose codimension is $\dim(N)$

Proof:
Apply the prior theorem and use the fact that $\operatorname{rank}(\Phi) = \dim(N)$. ■

Suppose M and N are C^k manifolds without boundaries and $\Phi : M \to N$ is a C^k map. We say a point $p \in M$ is a regular point of Φ if $\mathrm{d}\Phi_p$ is surjective. Otherwise we say p is a critical point of Φ. Also we say a point $c \in N$ is a regular value of Φ if $\Phi^{-1}(\{c\})$ if every $p \in \Phi^{-1}(\{c\})$ is a regular point of Φ. Otherwise, we say c is a critical value of Φ. Finally, a level set $\Phi^{-1}(\{c\})$ is a regular level set if c is a regular value of Φ.

As a side note, the set of regular points in M is always an open subset of U. This is because of proposition 4.1 on *pages 1054-1056*. So, we can always restrict the domain of Φ to just an open-submanifold where Φ is a submersion.

Corollary 5.14 (Regular level set Theorem): Suppose M and N are C^k manifolds without boundaries and $\Phi : M \to N$ is a C^k map. Every regular level set of Φ is a properly embedded C^k submanifold without a boundary and with codimension equal to $\dim(N)$.

Proof:
Let $c \in N$ be a regular value of Φ. The set $U \subseteq M$ of regular points of Φ is open in M and contains $\Phi^{-1}(\{c\})$ since c is a regular value. So, $\Phi|_U$ is a C^k submersion and the prior corollary shows that $\Phi^{-1}(\{c\})$ is a C^k submanifold of U without a boundary and with codimension equal to $\dim(N)$. Since the composition of two C^k embeddings is a C^k embedding, and the inclusion $\Phi^{-1}(\{c\}) \hookrightarrow M$ is the composition of the inclusions $\Phi^{-1}(\{c\}) \hookrightarrow U$ and $U \hookrightarrow M$, we can also conclude $\Phi^{-1}(\{c\})$ is an embedded C^k submanifold of M.

Also once again, Φ being continuous and $\{c\}$ being a closed subset of N means that $\Phi^{-1}(\{c\})$ is a closed subset of M. So, $\Phi^{-1}(\{c\})$ is a properly embedded submanifold. ■

Proposition 5.16: Let S be a subset of a C^r m-dimensional manifold M without a boundary. Then S is a k-dimensional embedded submanifold of M without a boundary if and only if every $p \in S$ has an open neighborhood $U \subseteq M$ such that $U \cap S$ is a zero set of a C^r submersion $\Phi : U \to \mathbb{R}^{m-k}$.

$(\Longrightarrow)$

Consider any $p \in S$ and let (U, φ) be a slice chart for S in M containing p. By translating the codomain of φ, we can without loss of generality assume that:

$$\varphi(U \cap S) := \{(x_1, \ldots, x_m) \in \varphi(U) : x_{k+1} = 0, \ldots, x_m = 0\}.$$

Now let us write $\varphi(q) = (x_1, \ldots, x_m)$ for all $q \in U$. Next define $\pi : \mathbb{R}^m \to \mathbb{R}^{m-k}$ by $(x_1, \ldots, x_m) \mapsto (x_{k+1}, \ldots, x_m)$. And finally, define $\Phi := \pi \circ \varphi$. Since both π and φ are C^r submersions, we can conclude that $\Phi : U \to \mathbb{R}^{m-k}$ is also a C^r submersion. Also, $\Phi^{-1}(\{(0, \ldots, 0)\}) = U \cap S$.

$(\Longleftarrow)$

By the submersion level set theorem, we can conclude that for every $p \in S$ there is an open set $U \subseteq M$ containing p such that $S \cap U$ is a k-dimensional C^r embedded submanifold of M without a boundary. And now we just apply the comment on *page 1116*. ■

Up until now, we've worked with submanifolds equipped with the subspace topology. However, this can be limiting. So, we adopt the following terminology.

Let M be a C^r manifold. A C^r immersed submanifold of M is a subset $S \subseteq M$ endowed with some topology (not necessarily the subspace topology) turning it into a topological manifold, as well a C^r structure such that the inclusion map $\iota : S \hookrightarrow M$ is a C^r immersion.

Remarks:

- If $k \leq r$ then we can define immersed C^k submanifolds of M identically as on *page 1111*.
- Since all embedded submanifolds are also immersed submanifolds, we take the convention that a "C^r submanifold" of M is only understood by default to be immersed. If we want that submanifold to also be embedded then we will add that adjective.
- The codimension of a C^r submanifold S of M is still defined as $\dim(M) - \dim(S)$. And S is a C^r hypersurface if S has codimension 1.

Proposition 5.18: Suppose M and N are C^r manifolds, and $F : N \to M$ is an injective C^r immersion. Also let $S := F(N)$. Then S has a unique topology and C^r structure making S into a C^r submanifold of M such that F is a C^r-diffeomorphism from N to S.

Proof:

To start off, we define a topology on S by saying that $U \subseteq S$ is open if $F^{-1}(U)$ is an open subset of N.

Note that this is a well-defined topology. After all, $F^{-1}(S) = N$ and $F^{-1}(\emptyset) = \emptyset$ are both open in N and hence S and $\emptyset$ are open subsets of S. Also suppose $\{U_\alpha\}_{\alpha \in A}$ is a collection of subsets of S such that $F^{-1}(U_\alpha)$ is open in N for all α. Then $F^{-1}(\bigcup_{\alpha \in A} U_\alpha) = \bigcup_{\alpha \in A} F^{-1}(U_\alpha)$ is open in N. Similar reasoning shows the topology on S is closed under finite intersections.

Importantly, this topology on S is the *unique* topology S must be equipped with in order for F to be a homeomorphism from N to S (which is necessary if we want F to also be a C^r-diffeomorphism).

Now N being second countable and Hausdorff means that S is as well. Plus we can equip S with an atlas given by the charts $(F(U), \varphi \circ F^{-1})$ where (U, φ) is any chart on N. By identical reasoning as in the proof of proposition 5.2 on *pages 1111-1112*, this atlas defines a C^r structure on S such that S is a topological manifold and $F : N \to S$ is a C^r-diffeomorphism. (In fact this atlas is itself maximal and F being a C^r-diffeomorphism from N to S would automatically imply S is equipped with this maximal atlas).

Finally, the inclusion map $S \hookrightarrow M$ is the composition of the diffeomorphism $F^{-1} : S \to N$ and the immersion $F : N \to M$. So, $S \hookrightarrow M$ is an immersion. ■

Proposition 5.22: If M is a C^r manifold and $S \subseteq M$ is a C^r (immersed) submanifold without corners, then for every $p \in S$ there is an open set $U \subseteq S$ containing p such that U is a C^r embedded submanifold of M.

Proof:
By the local embedding theorem on *pages 1105-1106*, since the inclusion $\iota : S \hookrightarrow M$ is an immersion we know every $p \in S$ has an open neighborhood $U \subseteq S$ such that $\iota|_U$ is an embedding. And then the image of an embedding is always an embedded submanifold. ■

As a side note, the prior proposition does *not* say that we can find an open set U in the topology of M containing p such that $U \cap S$ is an embedded submanifold. It only says we can find an open set U in the topology of S containing p such that U is an embedded submanifold.

Before talking about parametrizations of submanifolds, I want to skip ahead to some results which I think are more interesting yet still important. Firstly, we will establish two results about restricting the domain or codomain of a C^r map.

Theorem 5.27: Suppose M and N are C^r manifolds, $F : M \to N$ is a C^k map, and $S \subseteq M$ is an (immersed) C^r submanifold. Then $F|_S : S \to N$ is a C^k map.

Proof:
$F|_S = F \circ \iota$ where $\iota : S \hookrightarrow M$ is the C^r inclusion map. ■

Theorem 5.29: Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an (immersed) C^r submanifold without a boundary, and N is another C^r manifold. Next suppose $F : N \to M$ is a C^k map such that $F(N) \subseteq S$. If F is continuous as a map from N to S then $F : N \to S$ is a C^k map.

Proof:
Let $p \in N$ be arbitrary and set $q = F(p) \in S$. By proposition 5.22, we know there is an open neighborhood $V \subseteq S$ of q such that the inclusion map $\iota|_V : V \hookrightarrow M$ is a C^r embedding. But now V (equipped with the open submanifold structure relative to S) is an embedded C^r submanifold of M without a boundary. So, we let $\pi : \mathbb{R}^{\dim(M)} \to \mathbb{R}^{\dim(S)}$ be the projection map onto the first $\dim(S)$ many coordinates. Then by the comment on *page 1115* we can find a smooth chart (W, ψ) on M containing q such that $(W \cap V, \pi \circ \psi|_V)$ is a smooth chart on V containing q. Since every smooth chart on an open submanifold is also a smooth chart on the original manifold, we can further conclude $(W \cap V, \pi \circ \psi|_V)$ is a smooth chart on S containing q.

Now since $F : N \to S$ is continuous, we know that $U := F^{-1}(W \cap V)$ is an open subset of N containing p. So, we let (U_0, φ) be any smooth chart on N such that $p \in U_0 \subseteq U$. And now $(\pi \circ \psi|_V) \circ F \circ \varphi^{-1} = \pi \circ (\psi \circ F \circ \varphi^{-1})$ will be a C^k map since $F : N \to M$ is a C^k map.

This finishes proving that $F : N \to S$ is a C^k map. ■

Corollary 5.30 (The Embedded Case): Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an embedded C^r submanifold, and N is another C^r manifold. If $F : N \to M$ is a C^k map with $F(N) \subseteq S$ then $F : N \to S$ is also a C^k map.

Proof:
Since S is equipped with the subspace topology, we automatically have that $F : N \to M$ being continuous means that $F : N \to S$ is continuous. So, we can apply the prior theorem. ■

Suppose M is a C^r manifold and $S \subseteq M$ is an immersed C^r submanifold of M. We say S is weakly embedded in M and also an initial submanifold of M if every C^k map $F : N \to M$ (where $k \leq r$) whose image lies in S is also a C^k map from N into S. The prior corollary says that every embedded C^r submanifold is also a weakly embedded C^r submanifold.

A particularly important consequence of the prior corollary can be stated as follows:

Theorem 5.31: Suppose M is a C^r manifold without a boundary and $S \subseteq M$ is a weakly embedded C^r submanifold without a boundary. Then there is no other topology and C^r structure that S can be equipped with such that S is an immersed C^r submanifold of M without a boundary.

Proof:
Suppose $\widetilde{S}$ is the set S equipped with some other topology and C^r structure such that $\widetilde{S}$ is an immersed C^r submanifold of M without a boundary. Also let ι denote the inclusion map $S \hookrightarrow M$ and $\widetilde{\iota}$ denote inclusion map $\widetilde{S} \hookrightarrow M$.

Because $\widetilde{\iota}(S)$ is contained in S which is weakly embedded, we know $\widetilde{\iota}$ is a C^r map from $\widetilde{S}$ into S. Yet then note for any $p \in S$ that $\mathrm{d}\widetilde{\iota}_p : T_p\widetilde{S} \to T_pM$ is both injective and the composition of the maps:

$$T_p\widetilde{S} \xrightarrow{\mathrm{d}\widetilde{\iota}_p} T_pS \xrightarrow{\mathrm{d}\iota_p} T_pM$$

Thus we must have that $\widetilde{\iota}$ is a C^r immersion from $\widetilde{S}$ into S. In other words, $\widetilde{\iota}$ has constant rank. And since $\widetilde{\iota}$ is a bijection between S and $\widetilde{S}$, we can then conclude by the corollary on *page 1066* that $\widetilde{\iota} : \widetilde{S} \to S$ is a C^r-diffeomorphism. This implies that the topology and C^r structure on S and $\widetilde{S}$ are the same. ■

In particular, the prior theorem tells us that there is only one C^r structure that an embedded submanifold S of M (without a boundary) can be equipped with. So, the C^r structure constructed in theorem 5.8 based on the local k-slice condition (see *pages 1114-1116*) is the unique structure on S.

For a general immersed submanifold of M though, we have the following weaker uniqueness result.

Theorem 5.32: Suppose M is a C^r manifold without a boundary and $S \subseteq M$ is an immersed C^r submanifold without a boundary. Then with the given topology on S there is no other C^r structure that S can be equipped with such that S is an immersed C^r submanifold of M without a boundary.

Proof:
Let $\widetilde{S}$ be the set S equipped with the same topology but just a different C^r structure. Then the proof of theorem 5.31 works entirely the same, except that instead of using corollary 5.30, we instead note that $\widetilde{\iota}$ is a continuous function from $\widetilde{S}$ into S and so theorem 5.29 applies. ■

8/31/2026

I could allow for more types of corner charts and then try to prove an immersion rank theorem for those manifolds but ehhh I don't think it would likely be the end of the world to just limit myself to having a rank theorem for boundary charts (non-corner charts) in the domain.

Let M be an n-dimensional C^r manifold without a boundary (where $n \geq 1$) and let (U, φ) be a smooth chart on M. Given an integer $k \in \{0, 1, \ldots, n\}$, we say a k-dimensional half slice of U is a set $V \subseteq U$ such that:

$$\varphi(V) = \{(x_1, \ldots, x_n) \in \varphi(U) : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\},$$

where $c_{k+1}, \ldots, c_n \in \mathbb{R}$.

We say a subset $S \subseteq M$ satisfies the local k-slice condition for a manifold with a boundary (what a mouthful) if every $p \in S$ is contained in a smooth chart (U, φ) such that $S \cap U$ is either a k-slice or a k-half slice of U. In the former case, (U, φ) is called an interior slice chart for S in M. Meanwhile, in the latter case (U, φ) is called a boundary slice chart for S in M.

By using the local immersion theorem for manifolds with boundaries (see *pages 1101-1102*) in place of the rank theorem, we can prove an analogous result to theorem 5.8 on *pages 1114-1116*. Then in turn, we can also prove an analogous result to theorem 5.29. I shall write those results right belows. There's no new ideas in proving them though.

Theorem 5.51: Let M be a C^r n-dimensional manifold without a boundary and let $S \subseteq M$ be a C^r embedded k-dimensional submanifold without corners. Then S satisfies the local k-slice condition for a manifold with a boundary.

In particular, let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection map onto the first k-coordinates. Then for any $p \in S$ we can find a boundary or interior slice chart (U, φ) on M such that $(S \cap U, \pi \circ \varphi)$ is a smooth chart on S containing p.

Conversely, if $S \subseteq M$ satisfies the local k-slice condition for a manifold with a boundary, then S equipped with the subspace topology is a topological manifold. Furthermore, it has a C^r structure making it into an embedded C^r submanifold in M without corners.

Specifically, we equip S with the structure generated by the charts $(U \cap S, \pi \circ \varphi)$ where (U, φ) is any boundary or interior slice chart for S and $\pi : \mathbb{R}^n \to \mathbb{R}^k$ is the projection map onto the first k-coordinates.

Theorem 5.53(b): Suppose M and N are C^r manifolds such that M has no boundary, and also suppose $S \subseteq M$ is an (immersed) C^r submanifold without corners. Now let $F : N \to M$ be a C^k map such that $F(N) \subseteq S$. If F is continuous as a function from N to S then $F : N \to S$ is a C^k map.

As a side note, theorem 5.51 is enough to tell us that there is at most one C^r structure on a set $S \subseteq M$ equipped with the subspace topology turning it into an embedded C^r submanifold of M without corners.

To see why, suppose $S \subseteq M$ is any subset equipped with a C^r structure such that it becomes a C^r embedded submanifold of M. Next let $\widetilde{S}$ be the subset S equipped with the k-slice C^r structure constructed on the converse direction of theorem 5.51. By the red text, we can see that the C^r structures of S and $\widetilde{S}$ both contain a common atlas of smooth charts. So, both structures must be identical. And since the C^r structure of $\widetilde{S}$ is defined entirely independently from the C^r structure on S, we can conclude that the C^r structure on $\widetilde{S}$ is the unique C^r structure turning S into an embedded C^r submanifold of M.

Now the connection between abstract manifolds and the definition of manifolds given on *pages 77-99* is that Munkres works with embedded submanifolds of $\mathbb{R}^n$ (equipped with the standard structure).

To see one direction of this equivalence, consider the following:

Note that Munkres' definition of a coordinate patch on a C^r manifold $S \subseteq \mathbb{R}^n$ is equivalent to that of a C^r embedding $\alpha : U \to \mathbb{R}^n$ where $U \subseteq \mathbb{R}^k$ (or H^k) is an open set and $\alpha(U) = V$ is an open subset of S (equipped with the subspace topoloy). Since every point is in the image of a coordinate patch, we can conclude there is a covering for S consisting of open sets $\{W_\alpha\}_{\alpha \in A}$ in M such that $W_\alpha \cap S$ is an embedded submanifold (without corners because none of the coordinate patches are from $\mathbb{R}^k_{\geq 0}$) for all α. It follows that S satisfies the local k-slice condition for a manifold with a boundary. So, S is an embedded C^r submanifold of $\mathbb{R}^n$.

Going the other way, I want to introduce parametrizations more generally.

Suppose M is a C^r manifold and $S \subseteq M$ is an immersed k-dimensional C^r submanifold of M. Then a local parametrization of S is a continuous map $X : U \to M$ (where $U \subseteq \mathbb{R}^k$, H^k, or $\mathbb{R}^k_{\geq 0}$ is an open set) whose image is an open subset of S and which is a homeomorphism when considered as a function from U to $X(U) \subseteq S$. If X is also a C^r diffeomorphism from U to $X(U) \subseteq S$, it is called a C^r parametrization. Meanwhile, if $X(U) = S$ then we say X is a global parametrization.

Proposition 5.23: Suppose M is a C^r manifold, $S \subseteq M$ is an immersed k-dimensional C^r submanifold of M, $\iota : S \hookrightarrow M$ is the inclusion map, and $U \subseteq \mathbb{R}^k$ (or H^k or $\mathbb{R}^k_{\geq 0}$) is an open set. A map $X : U \to M$ is a C^r parametrization if and only if there is a smooth coordinate chart (V, φ) on S such that $U = \varphi(V)$ and $X = \iota \circ \varphi^{-1}$.

$(\Longleftarrow)$
Let (V, φ) be a smooth chart on S such that $U = \varphi(V)$ and $X = \iota \circ \varphi^{-1}$. Then X is a continuous map with a continuous inverse φ from $X(U) = V \subseteq S$ to U. Also X is a C^r diffeomorphism from U to V.

$(\Longrightarrow)$
Let $X : U \to M$ be a C^r paremtrization. Then let $V = X(U)$ and $\varphi = X^{-1}$ (considered as a C^r map from $V \subseteq S$ to U). Clearly $X = \iota \circ \varphi^{-1}$ and $U = \varphi(V)$. Also I claim that (V, φ) is a smooth chart on M. After all, let (W, ψ) be any other smooth chart on M.

Since φ^{-1} is a C^r map by assumption, we know that $\mathrm{Id}_U \circ \varphi \circ \psi^{-1} = \varphi \circ \psi^{-1}$ is a C^r map. Similarly, as X is a C^r map from U into V, we know $\psi \circ X \circ \mathrm{Id}_U = \psi \circ \varphi^{-1}$ is a C^r map. ■

In particular, this proposition shows that every point in an immersed submanifold is contained in some parametrization.

9/5/2026

Augh never mind I want to broaden the corner charts I'm allowed to use.

Suppose M is an n-dimensional C^r manifold with corners and $p \in M$ is a corner point. Then given any chart (U, φ) on M containing p we must have that $\varphi(U)$ is open in $\mathbb{R}^n_{\geq 0}$ and that $\varphi(p) \coloneqq (x_1, \ldots, x_n)$ where two of the coordinates of $\varphi(p)$ are zero. In particular, suppose exactly k coordinates of $\varphi(p)$ are zero. By permuting those coordinates, we can

assume without loss of generality it is the first k coordinates of $\varphi(p)$ are zero. And then by translating the codomain of φ by the vector $(0,\ldots,0,x_{k+1},\ldots,x_n)$, we get a "chart" (U,φ') where $\varphi'(U)$ is an open subset of $[0,\infty)^k \times \mathbb{R}^{n-k}$ with $\varphi'(p)=0$. And this "chart" will be C^r compatible with the C^r structure on M.

Lemma: Suppose $k_1, k_2 < n$ with $k_1 \neq k_2$ and that $U \subseteq [0,\infty)^{k_1} \times \mathbb{R}^{n-k_1}$ and $V \subseteq [0,\infty)^{k_2} \times \mathbb{R}^{n-k_2}$ are open subsets of the respective subspace topologies which both contain 0. Then there is no C^1 diffeomorphism $\varphi : U \to V$ such that $\varphi(0)=0$.

Proof:
We basically proceed by identical reasoning as on *page 1041*. Without loss of generality we can assume $k_1 < k_2$. Then suppose for the sake of contradiction that such a map $\varphi : U \to V$ did exist.

Let $S := \{y=(y_1,\ldots,y_n) \in \mathbb{R}^n : y_1 = 0,\ldots,y_{k_1}=0\}$. Thus S is an $(n-k_1)$-dimensional vector subspace of $\mathbb{R}^n$. Next note that $\mathrm{D}\varphi(0)$ must be an invertible linear map (with the inverse map $\mathrm{D}\varphi^{-1}(0)$). Therefore, we can conclude that $T := (\mathrm{D}\varphi(0))(S)$ is also an $(n-k_1)$-dimensional vector subspace.

Yet then there must exist $v=(v_1,\ldots,v_n) \in T$ such that one of the $v_1,\ldots,v_{k_2}$ equals zero. Otherwise, T would be contained in an $(n-k_2)$-dimensional vector space. And since $k_1 < k_2$, this would contradict that $(n-k_1)$-dimensional vector subspace of $\mathbb{R}^n$.

By negating v if necessary we can assume $v_i < 0$ for some $1 \leq i \leq k_2$. Finally, let $\gamma : (-\varepsilon,\varepsilon) \to U \cap S$ be a C^1 curve such that $\gamma(0)=0$ and

$$(\varphi \circ \gamma)'(0) = (\mathrm{D}\varphi(0))\gamma'(0) = v.$$

Then for small enough $t \in (0,\varepsilon)$ we must have that $(\alpha \circ \gamma)(t) \cdot e_i < 0$. But this contradicts that $\varphi(U \cap S) \subseteq [0,\infty)^{k_2} \times \mathbb{R}^{n-k_2}$. ■

Given a smooth chart (W,ψ) about a corner point p in a C^r manifold M such that $\psi(W)$ is an open subset of $\mathbb{R}^n_{\geq 0}$, we say p has depth k if exactly k of the components of $\psi(p)$ are zero. The prior lemma lets us conclude that the depth of p is defined independently of which chart we choose.

That said, because I want to allow smooth charts (W,ψ) about a corner point $p \in M$ to be "centered" (i.e. satisfy that $\psi(p)=0$), I'm going to start letting corner charts map to $[0,\infty)^k \times \mathbb{R}^{n-k}$ instead of just $[0,\infty)^n$.

Importantly, once we allow these new charts then we can by identical reasoning prove that the local immersion theorem for manifolds with boundaries on *pages 1101-1102* holds for a C^r manifold M with corner points. Then in turn, the local embedding theorem on *pages 1105-1106* also holds if the C^r manifold M has corners.

9/6/2026

If M is a C^r manifold, we say $D \subseteq M$ is a regular domain in M if it is a properly embedded codimension 0 submanifold of M.

Proposition 5.46: Suppose M is a C^r manifold without a boundary and $D \subseteq M$ is a regular domain. Then the topological interior and boundary of D is equivalent to the manifold interior and boundary of D respectively.

Proof:
For the sake of notation, I will let ∂D refer to the manifold boundary of D and $\mathrm{Int}(D)$ refer to the manifold interior of D.

Suppose $p \in \partial D$ with depth $k \geq 1$. Then by the local immersion theorem for manifolds with boundaries, there exists a smooth chart (U, φ) centered at p in D and a smooth chart (V, ψ) centered at p in M such that $\varphi(p) = 0$ and $\psi \circ \varphi(x_1, \ldots, x_n) = (x_1, \ldots, x_n)$. (Here $n = \dim(M) = \dim(D)$).

But now since D has the subspace topology, we know there is an open set $W \subseteq M$ such that $U = W \cap D$. In turn, $V_0 := V \cap W$ is an open set in M such that $V_0 \cap D = \psi^{-1}([0, \infty)^k \times \mathbb{R}^{n-k})$. And since $\psi(p) = (0, \ldots, 0)$ and $\psi(V_0)$ is an open subset of $\mathbb{R}^n$ containing $\psi(p)$, we can conclude that any open subset of p will contain points inside and outside D. So, p is in the topological boundary of D.

Meanwhile, suppose p is in the manifold interior of D. Importantly, the manifold interior is an open submanifold of M (see *page 1111*). So, p is contained in an open subset of M which is contained in D. Thus p is in the topological interior of D.

Since the topological interior of D is contained in D we can now conclude that the topological interior of D is precisely $\mathrm{Int}(D)$. Meanwhile, because D is a closed subset of M (since it is properly embedded), we know that the topological boundary of D is also contained in D. So, it must be precisely ∂D. ■

Proposition 5.47: Suppose M is a C^r manifold without a boundary and $f \in C^r(M, \mathbb{R})$.

(a) For each regular value $b \in \mathbb{R}$ we have that $f^{-1}((-\infty, b])$ is a regular domain in M without corners.

Proof:
To start off, f being continuous implies that $D := f^{-1}((-\infty, b])$ will be a closed subset of M. Thus, once we show D is an embedded C^r submanifold of M, it will be obvious that D is properly embedded.

We can proceed by showing that if $p \in D$ then there is an open set $U \subseteq M$ containing p such that $D \cap U$ is an embedded submanifold of codimension 0. Fortunately, by the continuity of f, if $f(p) \neq b$ then $f^{-1}((-\infty, b))$ is an open subset of M and of D containing p. Since open subsets of M are submanifolds of codimension 0 without boundaries, we are done in that case.

(In fact this would show that $f^{-1}((-\infty, b)) \subseteq \mathrm{Int}(D)$.)

Meanwhile suppose $f(p) = b$. Since b is a regular value of f, we know that $\mathrm{d}f_p$ is surjective. In turn, by proposition 4.1 on *page 1054* we can find an open set $U \subseteq M$ containing p such that $\mathrm{d}f_q$ is surjective for all $q \in U$.

By the rank theorem, there's a smooth chart (V, φ) on U and a smooth chart (W, ψ) on $\mathbb{R}$ such that $\varphi(p) = 0$ and $\psi \circ f \circ \varphi^{-1}$ is equal to the projection map $\pi : \mathbb{R}^n \to \mathbb{R}$ onto the first coordinate. (Here $n = \dim(M)$.) Yet since $\psi(b) = 0$ and ψ must map connected components of $W - \{b\}$ to $\psi(W) - \{0\}$, we can conclude that $W \cap (-\infty, b]$ is precisely given by either $\psi^{-1}((-\infty, 0])$ or $\psi^{-1}([0, \infty))$. By negating components of φ and ψ if necessary, we can without loss of generality assume $W \cap (-\infty, b] = \psi^{-1}([0, \infty))$.

Now $q \in V \cap D$ iff $q \in V$ and $f(q) \leq b$. Yet $f(q) \leq b$ iff:

$$\psi(f(q)) = (\psi \circ f \circ \varphi^{-1})(\varphi(q)) \geq 0.$$

So $V \cap D = \varphi^{-1}([0, \infty) \times \mathbb{R}^{n-1})$. In other words, $V \cap D$ is an n-half slice of the smooth chart (V, φ) on M containing p. It follows that $D \cap V$ is an embedded submanifold of M of codimension 0.

(And in particular p is a boundary point. Thus $f^{-1}(\{b\}) = \partial D$).

(b) Given any two regular values $a, b \in \mathbb{R}$ with $a < b$, we have that $f^{-1}([a, b])$ is a regular domain in M without corners.

Proof:
The proof of part (a) translates basically verbatim. The only additional step is that you need to shrink the open set U containing $p \in f^{-1}(\{b\})$ by intersecting it with $f^{-1}(b - \varepsilon, b + \varepsilon)$ for some small enough $\varepsilon > 0$. And you'd do an analogous thing when showing the case for some $p \in f^{-1}(\{a\})$. ■

(In particular, if $D := f^{-1}([a, b])$ then $f^{-1}((a, b)) = \mathrm{Int}(D)$ and $f^{-1}(\{a, b\}) = \partial D$).

9/10/2026

Lemma 5.34: Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an (immersed) C^r submanifold, and $f \in C^r(S, \mathbb{R})$ (meaning f is a C^r map according to the differentiable structure on S).

(a) If S is embedded and without a boundary, then there exists an open set $U \subseteq M$ containing S and a function $\widetilde{f} \in C^r(U, \mathbb{R})$ such that $f = \widetilde{f}|_S$.

Proof:
First I shall show that for every $p \in S$ there is an open set $U_p \subseteq M$ and a function $f_p \in C^r(U_p, \mathbb{R})$ such that $f = f_p$ on $U \cap S_p$.

Let $k = \dim(S)$, $n = \dim(M)$, $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates. Then we can find a slice chart (U_p, φ) for S in M containing p. In particular, $(S \cap U_p, \pi \circ \varphi)$ is a smooth chart on S with $S \cap U_p = \varphi^{-1}(\{(x_1, \ldots, x_n) : x_{k+1} = c_{k+1}, \ldots, x_n = c_n\})$. Next, let $\iota : \mathbb{R}^k \to \mathbb{R}^n$ be the function $(x_1, \ldots, x_k) \mapsto (x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$.

Given the local coordinates defined by (V, φ) we want $f_p(x_1, \ldots, x_n) = f(x_1, \ldots, x_k)$. So, let:

$$f_p = f \circ \varphi^{-1} \circ \iota \circ \pi \circ \varphi$$

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Then $f_p \in C^r(U_p, \mathbb{R})$. Also for all $(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n) \in S \cap U_p$ we know that:

$$f_p \circ \varphi^{-1}(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n) = f \circ \varphi^{-1}(x_1, \ldots, x_k, c_{k+1}, \ldots, c_n)$$

So, $f = f_p$ on $S \cap U_p$.

Now for every $p \in S$ let $U_p \subseteq M$ be an open set containing p and $f_p \in C^r(U_p, \mathbb{R})$ be a function with $f = f_p$ on $U \cap S_p$. Next, let $\{\psi_p\}_{p \in S}$ be a C^r partition of unity on the open submanifold $U := \bigcup_{p \in S} U_p$ of M subordinate to the sets U_p. (In other words $\operatorname{supp}(\psi_p) \subseteq U_p$ for every p and $\sum_{p \in S} \psi_p \equiv 1$.)

Then set $\widetilde{f} = \sum_{p \in S} f_p \psi_p$. This will be a C^r map from the open set U containing S. After all, every $q \in U$ has a neighborhood in U where only finitely many ψ_p are nonzero. And then $\widetilde{f}$ is automatically a C^r function on that neighborhood. Plus, since $f_p = f$ on $S \cap \operatorname{supp}(\psi_p)$ for all $p \in S$, we have that $\widetilde{f} = f \cdot 1 = f$ on S.

(b) If S is properly embedded then we can take U to be all of M.

Proof:
Let $\{(U_p, \varphi_p)\}_{p \in S}$ be the open covering of S consisting of slice charts from part (a). Then also let $U = M - S$. Since S is closed, we know that U is an open set. And now, we let $\{\psi_p\}_{p \in S} \cup \{\psi\}$ be a partition of unity on M subordinate to those slice charts as well as U. In other words, $\operatorname{supp}(\psi_p) \subseteq U_p$ for all $p \in S$, $\operatorname{supp}(\psi) \subseteq U$, and $\psi + \sum_{p \in S} \psi_p \equiv 1$ on M.

Now set $\widetilde{f} = \sum_{p \in S} f_p \psi_p$. Then $\widetilde{f}$ will be a C^r map from M into $\mathbb{R}$ After all, every $q \in M$ has a neighborhood where only finitely many of the ψ_p and ψ are nonzero. Furthermore, since $\psi \equiv 0$ on S, we know that $\sum_{p \in S} \psi_p \equiv 1$ on S. So, $\widetilde{f} = f \cdot 1 = f$ on S. ■

I'd like to take a break from John Lee chapter 5 and introduce a different important topic in the theory of manifolds.

Let M be a topological space. A (real) vector bundle of rank k over M is a topological space E with a surjective continuous map $\pi : E \to M$ satisfying that:

(a) For each $p \in M$ the fiber $E_p := \pi^{-1}(\{p\})$ over p is endowed with a structure turning it into a k-dimensional real vector space.

(b) For each $p \in M$ there is an open neighborhood U of p in M and a homeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ (called a local trivialization of E over U) satisfying that:

- $\pi_U \circ \Phi = \pi$ (where $\pi_U : U \times \mathbb{R}^k \to U$ is the projection);
- For each $q \in U$ the restriction of Φ to E_q is a vector space isomorphism from E_q to $\{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

If E and M are C^r manifolds, $\pi : E \to M$ is a C^r map, and the local trivializations can be chosen to be C^r diffeomorphisms, then E is said to be a C^r vector bundle over M. Also, any local trivialization that is a C^r diffeomorphism is called a "smooth" local trivialization.

We call E the total space of the bundle, M the base space of the bundle, and $\pi : E \to M$ the projection of the bundle.

Exercise 10.1: Suppose $\pi : E \to M$ is a C^r vector bundle (of rank k). Then the projection map π is a C^r submersion.

Proof:
Let $q \in E$ be any point, and then set $p = \pi(q)$. We can find an open set $U \subseteq M$ containing p and a C^r diffeomorphism $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^k$ that is a local trivialization. Because Φ is continuous, we know $V := \pi^{-1}(U)$ is an open subset of E containing q.

But now note that $\pi|_V = \pi_U \circ \Phi$ where $\pi_U : U \times \mathbb{R}^k \to U$ is the projection map. Therefore, $\mathrm{d}\pi_q = \mathrm{d}(\pi_U)_{\Phi(q)} \circ \mathrm{d}\Phi_q$. Yet we can also see that π_U is a submersion. After all, for any $x \in U \times \mathbb{R}^k$ we have that:
$T_x(U \times \mathbb{R}^k) \cong T_{\pi_U(x)}U \oplus T_{\pi_{\mathbb{R}^k}(x)}\mathbb{R}^k$ via the map sending v to $(\mathrm{d}(\pi_U)_x v,\ \mathrm{d}(\pi_{\mathbb{R}^k})_x v)$.

Thus, $\mathrm{d}(\pi_U)_x$ must be a surjection onto $T_{\pi_U(q)}U$. And since x was arbitrary, we can conclude π_U is a submersion.

More generally, if $M_1, \ldots, M_k$ are any manifolds, $1 \leq i \leq k$, and $\pi : M_1 \times \cdots \times M_k \to M_i$ is the projection map, then this reasoning shows that π is a submersion.

Also $\mathrm{d}\Phi_q$ must be an invertible map since Φ is a diffeomorphism. So, $\mathrm{d}\pi_q$ is the composition of two surjective linear maps, and in turn must be a surjective linear map itself. Since q was arbitrary, this proves that $\pi : E \to M$ is a submersion. ■

Lemma 1.35: Let M be a set, $r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$ and suppose we are given a collection $\{U_\alpha\}_{\alpha \in A}$ of subsets of M together with maps $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ such that:

(i) For each $\alpha \in A$, φ_α is a bijection between U_α and an open subset $\varphi_\alpha(U_\alpha)$ of $[0, \infty)^k \times \mathbb{R}^{n-k}$ (where $0 \leq k \leq n$);

(ii) For every $\alpha, \beta \in A$ we have that $\varphi_\alpha(U_\alpha \cap U_\beta)$ is an open subset of $\varphi_\alpha(U_\alpha)$;

(iii) Whenever $U_\alpha \cap U_\beta \neq \emptyset$ then the map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is a C^r map;

(iv) Countably many of the U_α cover M;

(v) For any $p, q \in M$ either there exists U_α with both $p, q \in U_\alpha$ or there exists disjoint U_α and U_β with $p \in U_\alpha$ and $q \in U_\beta$.

Then M has a unique topology such that each $(U_\alpha, \varphi_\alpha)$ is a chart. And if $r \geq 1$ then M has a unique C^r structure such that each $(U_\alpha, \varphi_\alpha)$ is a smooth chart.

To start off, we equip M with the topology generated by the collection:

$$\{\varphi_\alpha^{-1}(V) : \alpha \in A \text{ and } V \text{ is an open subset of } \varphi_\alpha(U_\alpha)\}.$$

To prove that the latter set is a basis, first note by (iv) that every $x \in M$ is contained in some U_α. Also suppose that $x \in \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$. Then (iii) implies that:

$$(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W) \text{ is an open subset of } \varphi_\alpha(U_\alpha \cap U_\beta).$$

By (ii), we also know $\varphi_\alpha(U_\alpha \cap U_\beta)$ is an open subset of $\varphi_\alpha(U_\alpha)$. Thus, $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)$ is also an open subset of $\varphi_\alpha(U_\alpha)$. And now, we can say that:

$$\varphi_\alpha^{-1}(V \cap (\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W)) = \varphi_\alpha^{-1}(V) \cap \varphi_\beta^{-1}(W)$$

is also a basis set. Equality here is due to (i) saying φ_α and φ_β are bijections.

Having equipped M with this topology, it's now clear that each φ_α is a homeomorphism between U_α and $\varphi_\alpha(U_\alpha)$.

To see why, first note by the definition of the topology that each φ_α is continuous. I also claim each φ_α is an open map. After all, suppose $V \subseteq U_\alpha$ is any open set and $x \in V$ is any point. Then we know there exists some open set $B \coloneqq \varphi_\beta^{-1}(W)$ where W is an open subset of $\varphi_\beta(U_\beta)$ and $x \in \varphi_\beta^{-1}(W) \subseteq V$. Yet also note that as $B \subseteq V \subseteq U_\alpha$, we can further say that W is an open subset of $\varphi_\beta(U_\beta \cap U_\alpha)$. In turn, by (ii) and (iii) we know that $(\varphi_\beta \circ \varphi_\alpha^{-1})^{-1}(W) = \varphi_\alpha(B)$ is an open subset of $\varphi_\alpha(U_\alpha)$. And since $\varphi_\alpha(B) \subseteq \varphi_\alpha(V)$, we've proven that $\varphi_\alpha(x)$ is in the topological interior of $\varphi_\alpha(V)$. Yet x was arbitrary. So, $\varphi_\alpha(V)$ is an open subset of $\varphi_\alpha(U_\alpha)$.

It follows that M is locally Euclidean. Furthermore, (iv) guarentees that M is second countable. Also, (v) guarentees that M is Hausdorff. So, we've equipped M with a topology turning it into a topological n-dimensional manifold.

As a side note, this topology is the coarsest topology such that every φ_α is continuous. Also suppose $\mathcal{T}'$ is another topology on M turning it into a topological manifold with each $(U_\alpha, \varphi_\alpha)$ as a chart. Then suppose $V \in \mathcal{T}'$. Since each φ_α is a homeomorphism, we must have that $\varphi_\alpha(V \cap U_\alpha)$ is an open subset of $\varphi_\alpha(U_\alpha)$ for all α. Yet in turn $V \cap U_\alpha$ must be in our original topology. And since $V = V \cap M = V \cap \bigcup_{\alpha \in A} U_\alpha = \bigcup_{\alpha \in A}(V \cap U_\alpha)$, we can conclude that V is open in our original topology. So, our original topology is also the finest topology that works. And this proves the topology we equipped M with is unique.

Finally, if $r \geq 1$ then (iii) also tells us the $(U_\alpha, \varphi_\alpha)$ are C^r compatible. And since they cover all of M, we know that they define a unique C^r structure on M. ■

The prior lemma is useful for constructing C^r manifolds. Here's one other observation that's necessary for what I'm about to do.

Suppose M is a C^r manifold (where $r \geq 1$) and (U, φ) and (V, ψ) are smooth charts such that $U \cap V = \emptyset$. We'll write $\varphi(p) = (x_1(p), \ldots, x_n(p))$ and $\psi(p) = (\widetilde{x_1}(p), \ldots, \widetilde{x_n}(p))$ Then for any $p \in U \cap V$ we can write $v \in T_pM$ as either $\sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p$ or as $\sum_{i=1}^n \widetilde{c_i} \left.\frac{\partial}{\partial \widetilde{x_i}}\right|_p$ (where each representation is unique). Importantly though, by taking coordinate representations of the identity map Id_M, we can see that:

$$|\widetilde{c_1}, \ldots, \widetilde{c_n}\rangle = (\mathrm{D}(\psi \circ \varphi^{-1})(\varphi(p)))|c_1, \ldots, c_n\rangle.$$

The most important vector bundle is the tangent bundle TM of a C^r manifold M.

We'll define $TM := \coprod_{p \in M} T_pM = \{(p, v) : p \in M \text{ and } v \in T_pM\}$. Then we'll also define $\pi : TM \to M$ by $\pi(p, v) = p$.

Lemma 3.18: Suppose $r \geq 1$. For any C^r n-dimensional manifold M, TM has a natural topology (and C^{r-1} structure if $r \geq 2$) turning TM into a $2n$-dimensional C^{r-1} manifold. Also, $\pi : TM \to M$ is a C^{r-1} map.

Proof:
Let (U, φ) be any smooth chart and let $\varphi(p) = (x_1(p), \ldots, x_n(p))$ be the local coordinates on that chart. Then we define a map $\widetilde{\varphi} : \pi^{-1}(U) \to \varphi(U) \times \mathbb{R}^n$ by:

$$\widetilde{\varphi}(p, \sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) = (x_1(p), \ldots, x_n(p), c_1, \ldots, c_n).$$

This is a bijection to the open set $\varphi(U) \times \mathbb{R}^n$. So, we let $(\pi^{-1}(U), \widetilde{\varphi})$ be a chart. By constructing such charts for all smooth charts on M, we get a collection of smooth charts for which condition (i) in the prior lemma is satisfied.

Next suppose (U, φ) and (V, ψ) are smooth charts on M. Then we can easily see that

- $\widetilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \varphi(U \cap V) \times \mathbb{R}^n$;
- $\widetilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V)) = \psi(U \cap V) \times \mathbb{R}^n$.

The latter sets are all open so condition (ii) is satisfied.

Furthermore, $\widetilde{\psi} \circ \widetilde{\varphi}^{-1} : \widetilde{\varphi}(\pi^{-1}(U) \cap \pi^{-1}(V)) \to \widetilde{\psi}(\pi^{-1}(U) \cap \pi^{-1}(V))$ is given by the map:

$$((x_1, \ldots, x_n), (c_1, \ldots, c_n)) \mapsto (\psi \circ \varphi^{-1}(x_1, \ldots, x_n), (\mathrm{D}(\psi \circ \varphi^{-1})(\varphi(p)))|c_1, \ldots, c_n\rangle)$$

Since $\psi \circ \varphi^{-1}$ is a C^r map, $\widetilde{\psi} \circ \widetilde{\varphi}^{-1}$ is thus a C^{r-1} map. And this fulfills condition (iii).

To show condition (iv), note that a countable collection $\{(U_n, \varphi_n)\}_{n \in \mathbb{N}}$ of smooth charts covers M. In turn, we can see that $\{(\pi^{-1}(U_n), \widetilde{\varphi_n})\}$ is a countable collection of charts covering TM.

Finally, to show condition (v) just note that if $p, q \in TM$ are in the same fiber of π then there is a single chart $(\pi^{-1}(U), \widetilde{\varphi})$ containing p and q. Meanwhile, if $\pi(p) \neq \pi(q)$ then we can find disjoint charts (U, φ) and (V, ψ) on M such that $\pi(p) \in U$ and $\pi(q) \in V$. And then $\pi^{-1}(U)$ and $\pi^{-1}(V)$ are disjoint sets containing p and q respectively.

And now we just apply lemma 1.35 to get our topology and C^{r-1} structure on TM turning it into a $2n$-dimensional C^{r-1} manifold.

To see that $\pi : TM \to M$ is a C^{r-1} map, just note for any smooth chart (U, φ) on M that $\pi(\pi^{-1}(U)) = U$ (since π is surjective) and $\varphi \circ \pi \circ \widetilde{\varphi}^{-1} = \pi_U$ where π_U is the projection map $U \times \mathbb{R}^n \to U$. It follows that every $q \in TM$ has a neighborhood on which π is a C^{r-1} map. So, π is a C^{r-1} map.

Now I claim that $\pi : TM \to M$ (where TM is equipped with the topology described priorly) is a vector bundle of rank $n \coloneqq \dim(M)$ over M.

Proof:
For every $p \in M$ we have that $\pi^{-1}(\{p\}) \cong T_pM$. Thus every fiber is an n-dimensional real vector space.

Also for any $p \in M$ we can pick a smooth chart (U, φ) on M containing p. And then after writing $\varphi(q) = (x_1(q), \ldots, x_n(q))$, we define $\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^n$ by:

$$\Phi(q, \textstyle\sum_{i=1}^n c_i \left.\frac{\partial}{\partial x_i}\right|_p) = (q, (c_1, \ldots, c_n)).$$

To see that Φ is a local trivialization, first note that $\pi_U \circ \Phi = \pi$ (where $\pi_U : U \times \mathbb{R}^n \to U$ is the projection map). Also note for each $q \in U$ that $\Phi|_{T_qM}$ is a linear map from T_qM to $\mathbb{R}^n$. ■

Furthermore, if M is a C^r manifold where $r \geq 2$, then TM is a C^{r-1} vector bundle over M.

Proof:
It suffices to show that Φ from before is a C^{r-1} diffeomorphism. Fortunately, note that $\varphi^{-1} \times \mathrm{Id}_{\mathbb{R}^n} : \varphi(U) \times \mathbb{R}^n \to U \times \mathbb{R}^n$ is a C^r diffeomorphism. Also, recall that $\widetilde{\varphi} : \pi^{-1}(U) \to \varphi(U) \times \mathbb{R}^n$ given by $\widetilde{\varphi}(p, \sum_{i=1}^n \left.\frac{\partial}{\partial x_i}\right|_p) = (\varphi(p), (c_1, \ldots, c_n))$ is a smooth chart on TM (and thus a C^{r-1} diffeomorphism). And since $\Phi = (\varphi^{-1} \times \mathrm{Id}_{\mathbb{R}^n}) \circ \widetilde{\varphi}$, we know that Φ is a C^{r-1} diffeomorphism. ■

Given a C^k map $F : M \to N$ between C^r manifolds, we define a map $\mathrm{d}F : TM \to TN$ (called the global differential) by $\mathrm{d}F(p, v) \coloneqq (F(p), \mathrm{d}F_p(v))$.

Proposition 3.21: Suppose $k \geq 1$. Then $\mathrm{d}F$ is a C^{k-1} map.

Proof:
We'll let $\pi_1 : TM \to M$ and $\pi_2 : TN \to N$ denote the projection maps of our vector bundles. Now suppose $(p, v) \in TM$. Then we know there exists smooth charts (U, φ) on

M containing p and (V,ψ) on N containing $F(U)$. In turn, we consider the smooth charts $(\pi^{-1}(U),\widetilde{\varphi})$ on TM and $(\pi^{-1}(V),\widetilde{\psi})$ on TN (where $\widetilde{\varphi}$, $\widetilde{\psi}$ were constructed in lemma 3.18). These charts will satisfy that $(p,v)\in\pi^{-1}(U)$ and $\mathrm{d}F(\pi^{-1}(U))\subseteq\pi^{-1}(V)$.

But now let us write $\varphi(q)=(x_1(q),\ldots,x_m(q))$ and $\psi(q)=(y_1(q),\ldots,y_n(q))$. Also let us set $\widehat{F}\coloneqq\psi\circ F\circ\varphi^{-1}$. Then:

$$\widetilde{\psi}\circ\mathrm{d}F\circ\widetilde{\varphi}^{-1}((x_1,\ldots,x_m),(c_1,\ldots,c_m)) = (\widehat{F}(x_1,\ldots,x_m),\ (\mathrm{D}\widehat{F}(x_1,\ldots,x_m))|c_1,\ldots,c_m\rangle)$$

This proves that $\mathrm{d}F:TM\to TN$ is a C^{k-1} map. ■

As a side note, if $r\geq 2$ we can now define a functor $\mathbf{Diff}_c^r\to\mathbf{Diff}_c^{r-1}$ sending any manifold M to TM and any C^r map $F:M\to N$ to $\mathrm{d}F$. Meanwhile, if $r=1$ then we can similarly define a functor $\mathbf{Diff}_c^1\to\mathbf{Man}_b$. After all, because $\mathrm{d}(G\circ F)_p=\mathrm{d}G_{F(p)}\circ\mathrm{d}F_p$, we can easily conclude that $\mathrm{d}(G\circ F)=\mathrm{d}G\circ\mathrm{d}F$. Similarly, we can conclude $\mathrm{d}(\mathrm{Id}_M)=\mathrm{Id}_{TM}$. So, these functors are well-defined.

Typically, given the map $\pi:TM\to M$ we'll just identify T_pM with $\pi^{-1}(\{p\})$ for every $p\in M$. That way, we can formally write $\mathrm{d}F$ instead of $\mathrm{d}F_p$ if the point p is obvious from context.

Let $J\subseteq\mathbb{R}$ be some interval and suppose M is a C^r manifold. Given a C^1 map $\gamma:J\to M$ and any $t_0\in J$ we define the velocity of γ at t_0 to be $\gamma'(t_0)\coloneqq\mathrm{d}\gamma(\frac{\mathrm{d}}{\mathrm{d}t}\big|_{t_0})$. Some other common notations for $\gamma'(t_0)$ are $\dot{\gamma}(t_0)$ and $\frac{\mathrm{d}\gamma}{\mathrm{d}t}(t_0)$.

Note that if $f\in C^1(U,\mathbb{R})$ where U is an open set containing $\gamma(t_0)$ then:

$$\gamma'(t_0)f=\mathrm{d}\gamma\left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t_0}\right)f=\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t_0}(f\circ\gamma)=(f\circ\gamma)'(t_0)$$

Next let (U,φ) be a smooth chart on M containing $\gamma(t_0)$. Then by restricting to a small enough neighborhood of t_0 we can assume $\gamma(J)\subseteq U$. Thus, we can write $\varphi\circ\gamma(t)=(\gamma_1(t),\ldots,\gamma_n(t))$. And now in coordinates, we have that:

$$\gamma'(t_0)=\sum_{i=1}^n\frac{\mathrm{d}\gamma_i}{\mathrm{d}t}(t_0)\ \frac{\partial}{\partial x_i}\Big|_p$$

Proposition 3.23: Suppose M is a C^r manifold without corners and $p\in M$. Then every $v\in T_pM$ is the velocity of some C^r curve γ at 0.

Hopefully this is obvious from *pages 1046-1051* and *page 1057-1058*.

Now back to chapter 5.

Given an immersed C^r submanifold S of a C^r manifold M, we can view its tangent space T_pS at a point $p \in S$ as a subspace of the tangent space T_pM of M at p. To see how, recall by definition that the inclusion map $\iota : S \hookrightarrow M$ is an immersion. Thus at every $p \in S$ we have that $\mathrm{d}\iota_p : T_pS \to T_pM$ is an injection.

In particular, suppose $v \in T_pS$ and $\widetilde{v} = \mathrm{d}\iota_p v \in T_pM$. Then:

$$\widetilde{v}([f]) = \widetilde{v}([f, U]) = v([f \circ \iota, U \cap S]) = v([f|_S]).$$

Now we shall view the injection $\mathrm{d}\iota_p$ as an identification of T_pS as a subspace of T_pM.

Proposition 5.35: Suppose M is a C^r manifold, $S \subseteq M$ is an immersed C^r submanifold without corners, and $p \in S$. A vector $v \in T_pM$ is in T_pS if and only if there is a C^r curve $\gamma : J \to M$ such that:

- $J \subseteq \mathbb{R}$ is a compact interval containing 0,
- $\gamma(J) \subseteq S$,
- $\gamma(0) = p$,
- $\gamma'(0) = v$,
- γ is a C^r curve when viewed as a function from J into S.

$(\Longrightarrow)$
Suppose there exists $\widetilde{v} \in T_pS$ such that $v = \mathrm{d}\iota_p\widetilde{v}$. Then we know there is a C^r curve $\gamma : J \to S$ such that $J \subseteq \mathbb{R}$ is a compact interval containing 0, $\gamma(0) = p$, and $\gamma'(0) = \widetilde{v}$. But now $\iota \circ \gamma : J \to M$ is also a C^r curve (since it is a composition of two C^r functions). Also $(\iota \circ \gamma)'(0) = \mathrm{d}\iota_p \circ \mathrm{d}\gamma_0(\frac{\mathrm{d}}{\mathrm{d}t}\big|_0) = \mathrm{d}\iota_p(\gamma'(0)) = \mathrm{d}\iota_p(\widetilde{v}) = v$

$(\Longleftarrow)$
If we let $\widetilde{\gamma}$ be the curve γ with it's codomain restricted to S and then set $\widetilde{v} = \gamma'(0)$, we have that:

$$\mathrm{d}\iota_p(\widetilde{v}) = \mathrm{d}\iota_p \circ \mathrm{d}\widetilde{\gamma}_0(\tfrac{\mathrm{d}}{\mathrm{d}t}\big|_0) = \mathrm{d}(\iota \circ \widetilde{\gamma})_0(\tfrac{\mathrm{d}}{\mathrm{d}t}\big|_0) = (\iota \circ \widetilde{\gamma})'(0) = \gamma'(0) = v.$$

So, $v \in T_pM$ is also in T_pS. ■

Proposition 5.37: Suppose M is a C^r manifold without a boundary, $S \subseteq M$ is an embedded C^r submanifold (without corners), and $p \in S$. As a subspace of T_pM, we have that:

$$T_pS = \{v \in T_pM : v(f) = 0 \text{ if } f \in C^r(M, \mathbb{R}) \text{ with } f \equiv 0 \text{ on } S\}.$$

Proof:
First suppose $v \in T_pS \subseteq T_pM$. In other words, if $\iota : S \hookrightarrow M$ is the inclusion map then we know there exists $w \in T_pS$ such that $\mathrm{d}\iota_p(w) = v$. But now if $f \in C^r(M, \mathbb{R})$ is a function vanishing on S, we know that $f \circ \iota \equiv 0$. So:

$$v(f) = \mathrm{d}\iota_p(w)f = w(f \circ \iota) = w(0) = 0.$$

Conversely, suppose $v \in T_pM$ satisfies that $v(f) = 0$ for all $f \in C^r(M, \mathbb{R})$ with $f \equiv 0$ on S. Now we want to show that there is some $w \in T_pS$ with $v = \mathrm{d}\iota_p(w)$. Fortunately, we can pick some slice (or half slice) chart (U, φ) on M containing p such that:

$$U \cap S = \{\varphi^{-1}(x_1, \ldots, x_n) : (x_1, \ldots, x_n) \in \varphi(U) \text{ and } x_{k+1} = 0, \ldots, x_n = 0\}$$

If we let $\pi : \mathbb{R}^n \to \mathbb{R}^k$ be the projection onto the first k coordinates, we have that $(U \cap S, \pi \circ \varphi)$ is a smooth chart on S. But now with respect to this coordinate chart and the chart (U, φ) on M, we have that the coordinate representation of $\iota : S \hookrightarrow M$ is $\iota(x_1, \ldots, x_k) = (x_1, \ldots, x_k, 0, \ldots, 0)$. It then follows that if $\varphi(q) = (x_1(q), \ldots, x_n(q))$ then T_pS is the subspace of T_pM spanned by $\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_k}\Big|_p$.

Now write $v = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i}\Big|_p \in T_pM$. We just need to show that $c_j = 0$ for all $j \geq k+1$ in order to prove that $v \in T_pS$.

Fortunately, we can define a C^r function $f : M \to [0,1]$ such that $\operatorname{supp}(f) \subseteq U$ and $f \equiv 1$ on a neighborhood of p in M. Next given any given any $j \geq k+1$ we can define a C^r map $g : M \to \mathbb{R}$ by $g(q) = f(q) \cdot x_j(q)$. Then:

$$v(g) = v(q \mapsto x_j(q)) = c_j.$$

Yet we also know that $g \equiv 0$ on U^{C} and $g \equiv 0$ on $U \cap S$. So, $g \equiv 0$ on S and in turn $v(g) = 0$ by assumption. So, $c_j = 0$ for all $j \geq k+1$. ■

If $S \subseteq M$ is an embedded C^r submanifold of a manifold M, we say a C^r map $\Phi : M \to N$ such that S is a regular level set of Φ is a defining map for S. If $N = \mathbb{R}^{\dim(M)-\dim(S)}$, we also call Φ a defining function.

Also suppose $U \subseteq M$ is an open subset and $\Phi : U \to N$ is a C^r map such that $S \cap U$ is a regular level set of Φ. Then Φ is called a local defining map / function for S. Note that proposition 5.16 on *page 1120* guarentees that all embedded submanifolds without boundaries admit local defining functions on neighborhoods around all their points.

Proposition 5.38: Suppose M is a C^r manifold without a boundary and $S \subseteq M$ is an embedded submanifold without a boundary. If $\Phi : U \to N$ is a local defining map for S then $T_pS = \ker(\mathrm{d}\Phi_p)$ for all $p \in S \cap U$.

Proof:
Let $\iota : S \cap U \hookrightarrow M$ be the inclusion map and recall that T_pS is identified with the subspace $\operatorname{im}(\mathrm{d}\iota_p)$ of T_pM. But also note that $\Phi \circ \iota$ is a constant map. Therefore, $\mathrm{d}\Phi_p \circ \mathrm{d}\iota_p$ will be the zero map. And this proves that $\operatorname{im}(\mathrm{d}\iota_p) \subseteq \ker(\mathrm{d}\Phi_p)$.

On the other hand, since $\mathrm{d}\Phi_p$ is surjective, the rank-nullity theorem tells us that $\dim(\ker(\mathrm{d}\Phi_p)) = \dim(T_pM) - \dim(T_{\Phi(p)}N) = \dim(T_pS) = \dim(\operatorname{im}(\mathrm{d}\iota_p))$. So, we can actually conclude that $\operatorname{im}(\mathrm{d}\iota_p) = \ker(\mathrm{d}\Phi_p)$. ■

9/14/2026

Problem 5-3: Suppose M is a C^r manifold and $S \subseteq M$ is an immersed C^r submanifold. If any of the following conditions holds then S is an embedded submanifold.

(a) S has no boundary and has codimension 0 in M.

(b) The inclusion map $\iota : S \hookrightarrow M$ is proper.

(c) S is compact.

Proof:
In all cases, we need to show that $\iota : S \hookrightarrow M$ is a topological embedding. After all, that is equivalent to saying that S is equipped with the subspace topology relative to M.

For condition (a), note that ι being full rank actually let's us conclude by the inverse function theorem and by exercise 4-2 on *pages 1056-1057* that $S \subseteq \text{Int}(M)$ and ι is a local C^r diffeomorphism. In turn, as ι is also injective we know that it is a topological embedding.

For condition (b), recall that any continuous proper map into an LCH space is a closed map. Therefore, ι is a closed map. And since any injective continuous closed map is a topological embedding, we can conclude ι is a topological embedding.

For condition (c), note that since S is compact and M is Hausdorff, we can conclude that $\iota : S \hookrightarrow M$ is a closed map. And since ι is a continuous injection, we can conclude it is a topological embedding. ■

Problem 5-7: Let $F : \mathbb{R}^2 \to \mathbb{R}$ be defined by $F(x, y) = x^3 + xy + y^3$. Which level sets of F are smooth embedded submanifolds of $\mathbb{R}^2$?

To start off, $DF(x, y) = [3x^2 + y, 3y^2 + x]$. Thus, $(x, y) \in \mathbb{R}^2$ is a critical point of F iff $3x^2 + y = 0 = 3y^2 + x$. By solving this simple system of equations, we can in turn see the only critical points in $\mathbb{R}^2$ are $(0, 0)$ and $(-\frac{1}{3}, -\frac{1}{3})$. Since $F(0, 0) = 0$ and $F(-\frac{1}{3}, -\frac{1}{3}) = \frac{1}{27}$, we can conclude that if $c \notin \{0, \frac{1}{27}\}$ then $F^{-1}(\{c\})$ is a regular level set and in turn an embedded C^∞ submanifold without a boundary.

This leaves us with the task of concluding whether $F^{-1}(\{0\})$ or $F^{-1}(\{\frac{1}{27}\})$ are embedded C^∞ submanifolds.

- I claim $F^{-1}(\{\frac{1}{27}\})$ is not an embedded C^∞ submanifold. To see why, note that:

$$x^3 + xy + y^3 - \tfrac{1}{27} = x^3 + y^3 + (-\tfrac{1}{3})^3 - 3(\tfrac{-1}{3}xy)$$

Lemma: $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc)$.

Motivation:
Substitute in $a = -(b + c)$. Then:

$$a^3 + b^3 + c^3 - 3abc = -(b^3 + 3b^2c + 3bc^2 + c^3) + b^3 + c^3 + 3(b + c)bc = 0.$$

It follows that if $a+b+c = 0$ then $a^3+b^3+c^3-3abc = 0$. Or in other words, $(a + b + c)$ should be a factor of the polynomial $a^3 + b^3 + c^3 - 3abc$.

Next note that if $(a + b + c) \cdot P(a, b, c) = a^3 + b^3 + c^3 - 3abc$ then we can make the reasonable guess that P is a symmetric polynomial due to the fact that both other polynomials in the equation are symmetric. Also, since every term in $a + b + c$ has degree 1 and every term in $a^3 + b^3 + c^3 - 3abc$ has degree 3, we should have that every term of P has degree 2. So, we can assume $P(a, b, c)$ has the form:

$$P(a, b, c) = A(a^2 + b^2 + c^2) + B(ab + ac + bc)$$

By matching terms, we can see that $A = 1$ and $B = -1$. And now we just hope that this works.

Proof:

$$\begin{aligned}(a+b+c)(a^2+b^2+c^2-ab-ac-bc) &= a^3+ab^2+ac^2-a^2b-a^2c-abc \\ &\quad +a^2b+b^3+bc^2-ab^2-abc-b^2c \\ &\quad +a^2c+b^2c+c^3-abc-ac^2-bc^2 \\ &= a^3+b^3+c^3-3abc.\end{aligned}$$

Based on the prior lemma, we can now say that:

$$x^3+y^3+(-\tfrac{1}{3})^3-3(\tfrac{-1}{3}xy)=(x+y-\tfrac{1}{3})(x^2+y^2+\tfrac{1}{9}-xy+\tfrac{1}{3}(x+y))$$

Our level set is thus the line drawn by $y=-x+\frac{1}{3}$ unioned with whatever the zero set is of the function $G(x,y)=x^2+y^2+\frac{1}{9}-xy+\frac{1}{3}(x+y)$. To calculate this set, note that:

$$\begin{aligned}2G(x,y) &= x^2+x^2+y^2+y^2-2xy+\tfrac{1}{9}+\tfrac{1}{9}+2\tfrac{1}{3}x+2\tfrac{1}{3}y \\ &= (x-y)^2+(x+\tfrac{1}{3})^2+(y+\tfrac{1}{3})^2.\end{aligned}$$

Since squares are always nonnegative and scaling a function shouldn't change its zero set, we can conclude that $G(x,y)=0$ iff $x=y$, $x=-\frac{1}{3}$, and $y=-\frac{1}{3}$. So, we've proven that:

$$F^{-1}(\{\tfrac{1}{27}\})=\{(-\tfrac{1}{3},-\tfrac{1}{3})\}\cup\{(x,y):y=-x+\tfrac{1}{3}\}$$

This is not an embedded submanifold because it is a union of a 0-dimensional embedded submanifold and 1-dimensional embedded submanifold which are both disjoint.

- Next I claim $F^{-1}(\{0\})$ is also not an embedded C^∞ submanifold. To see why, consider that we can parametrize $F^{-1}(\{0\})$ as follows.

  Suppose $x\neq 0$. Then for any $(x,y)\in F^{-1}(\{0\})$ there exists some scalar $t\in\mathbb{R}$ such that $y=tx$. And now $x^3+tx^2+t^3x^3=0$. Since $x\neq 0$, we can divide by it to get that $x(1+t^3)+t=0$. So, $x=\frac{-t}{1+t^3}$. Next note that t can't equal -1 since that would give a contradiction that $t=0$. So, we can safely rearrange our equation to get that $x=\frac{-t}{1+t^3}$. And in turn, we can solve that $y=\frac{-t^2}{1+t^3}$.

  Meanwhile, note that $F(0,y)=y^3=0$ iff $y=0$. But then $(0,0)=(\frac{-0}{1+0^3},\frac{-0^2}{1+0^3})$. So, we can conclude that:

  $$F^{-1}(\{0\})\subseteq\{(\tfrac{-t}{1+t^3},\tfrac{-t^2}{1+t^3}):t\in\mathbb{R}-\{-1\}\}$$

  Checking the other inclusion is also now easy. So, we have a parametrization for $F^{-1}(\{0\})$.

  But now to finish off, note that as $t\to 0$ or ∞ or $-\infty$, we have that our parametrization approaches the origin from different directions. This would imply $T_0(F^{-1}(\{0\}))=T_0\mathbb{R}^2$ and in turn $F^{-1}(\{0\})$ is a 2-dimensional embedded submanifold. Yet we know that every other point of $F^{-1}(\{0\})$ has a neighborhood on which $F^{-1}(\{0\})$ is a 1-dimensional embedded submanifold. So, we can't have that $F^{-1}(\{0\})$ is an embedded submanifold. ■

I'm going to switch to chapter 6 of John Lee now. Note that we consider any measurable subset of $\mathbb{R}^n$ (for any n) as being equipped with the complete Lebesgue measure (which I'll denote μ).

Lemma 6.3: Suppose $n \geq 2$, A is an open or closed subset of $\mathbb{R}^{n-1}$ (or $[0,\infty)^k \times \mathbb{R}^{n-k-1}$ or $[0,\infty)^{n-1}$), and $f : A \to \mathbb{R}$ is a continuous function. Then the graph:

$$\Gamma(f) \coloneqq \{(x_1, \ldots, x_{n-1}, f(x_1, \ldots, x_n)) \in \mathbb{R}^n : (x_1, \ldots, x_{n-1} \in A)\}$$

is a Borel set of measure zero in $\mathbb{R}^n$.

Proof:

To start off $\Gamma(f)$ is a Borel measurable set. To see why, let $g : A \to A \times \mathbb{R}$ be defined by $g(x) = (x, f(x))$. This will be continuous function.

Next I claim that in the case that either A is open or closed, we can write A as a union of countably many compact sets $\{F_k\}_{k\in\mathbb{N}}$.

If A is closed then we merely intersect A with parallelotopes in a lattice to get our compact sets. Meanwhile, if A is open then we know A is an LCH space. So, every $x \in A$ has a compact neighborhood in A. And then since A is second countable, we can just take a countable subcover.

But now since g is continuous, we then know each $g(F_k)$ is a compact subset of $A \times \mathbb{R}$ (and in turn of $\mathbb{R}^n$). So, each $g(F_k)$ is a measurable set and $g(A) = \bigcup_{k\in\mathbb{N}} g(F_k)$. So, $\Gamma(f) = g(A)$ is a measurable set.

Now to show that $\Gamma(f)$ has measure zero, we just note by Fubini's theorem that:

$$\mu(\Gamma(f)) = \int_A \mu(\{f(x)\})\mathrm{d}x = \int_A 0\mathrm{d}x = 0. \blacksquare$$

Propositon 6.5: Suppose $U \subseteq \mathbb{R}^n$ is an open set and $f : U \to \mathbb{R}^n$ is a C^1 map. If $A \subseteq U$ is a set of Lebesgue measure zero then so is $f(A)$.

Proof:

By Corollary C.30 on *page 292*, we know that f is locally Lipschitz continuous. So every $x \in A$ has an open neighborhood N_x on which F is Lipschitz continuous. Yet because U is second countable, we then can pick countably many of those neighborhoods $\{N_{x_k}\}_{k\in\mathbb{N}}$ to cover A.

Next, write $A = \bigcup_{k\in\mathbb{N}}(A \cap N_{x_k})$. Since $A \cap N_{x_k} \subseteq A$ we know that it is Lebesgue measurable with measure zero. And since f is Lipschitz continuous on N_{x_k}, we can conclude by the lemma on *page 1009* that $f(A \cap N_{X_k})$ is also Lebesgue measurable with measure zero.

Finally, since $f(A) = \bigcup_{k\in\mathbb{N}} f(A_k)$ we can conclude that $f(A)$ is a Lebesgue measurable. Also $\mu(f(A)) \leq \sum_{k\in\mathbb{N}} \mu(f(A \cap N_{x_k})) = \sum_{k\in\mathbb{N}} 0 = 0$. So, $f(A)$ has measure zero. $\blacksquare$

As a side note, it is now also an easy corollary that if $U \subseteq [0,\infty)^k \times \mathbb{R}^{n-k}$ is an open set, $f : U \to \mathbb{R}^n$ is a C^1 map, and $A \subseteq U$ is a set of measure zero, then so is $f(A)$. After all, by definition f can be extended to a C^1 map to an open set $\widetilde{U} \subseteq \mathbb{R}^n$ containing U. And then we just apply the proposition from before.

Now for the time being, I don't have the background to define full on measures on abstract manifolds. However, it will still be useful to be able to classify if a subset of a manifold is negligable or not. Later on when I learn proper measure theory for manifolds, I can tie this content to that.

Let M be a C^r n-dimensional manifold (where $r \geq 1$ and $n \geq 1$). We can equip M with it's Borel σ-algebra $\mathcal{B}_M$.

We say $A \subseteq M$ has measure zero in M if there exists a Borel set $B \subseteq M$ such that $A \subseteq B$ and for every smooth chart (U, φ) on M we have that $\varphi(B \cap U)$ is a subset of $\mathbb{R}^n$ with Lebesgue measure zero.

As a side note, since $U \subseteq M$ is an open set and φ is a homeomorphism from U to a measurable subset of $\mathbb{R}^n$, we can guarentee that $\varphi(E \cap U)$ will be Borel measurable in $\mathbb{R}^n$ for every Borel measurable set $E \in \mathcal{B}_M$.

Remark:
It's easy to see that if $\mathcal{N}$ is the collection of sets of measure zero in M, then:

- $\emptyset \in \mathcal{N}$,
- $A \in \mathcal{N}$ and $A' \subseteq A \implies A' \in \mathcal{N}$,
- $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{N} \implies \bigcup_{k \in \mathbb{N}} A_k \in \mathcal{N}$.

The following two lemmas prove almost converse statements.

Lemma: If $A \subseteq M$ is a set of measure zero in M and (U, φ) is any smooth chart, then $\varphi(A \cap U)$ is Lebesgue measurable with measure zero.

Proof:
Since A has measure zero in M, we can find a Borel set $B \in \mathcal{B}_M$ such that $A \subseteq B$ and $\varphi(B \cap U) = 0$. Then as $\varphi(A \cap U) \subseteq \varphi(B \cap U)$, we know by the completeness of the Lebesgue measure that $\varphi(A \cap U)$ is Lebesgue measurable and a null set. ■

Lemma 6.6: Suppose $A \subseteq M$ and that $\{(U_i, \varphi_i)\}_{i \in I}$ is a collection of smooth charts covering A such that $\varphi_i(A \cap U_i)$ is Lebesgue measurable with measure zero for all $i \in I$. Then A has measure zero in M.

Proof:
Since M is second countable, we can without loss of generality assume our indexing set I equals $\mathbb{N}$ by taking a countable subcover of A. Then note that $A = \bigcup_{i \in \mathbb{N}} (A \cap U_i)$. Next, since $\varphi(A \cap U_i)$ is Lebesgue measurable with measure zero for every i by assumption, we know there is a Borel measurable set $E_i \subseteq \varphi(U_i)$ of measure zero containing $\varphi(A \cap U_i)$. And now we set $B_i := \varphi^{-1}(E_i)$. This will be a Borel subset of M (since φ is continuous.) Also, $A \cap U_i \subseteq B_i \subseteq U_i$ for each i.

Let $B := \bigcup_{i \in \mathbb{N}} B_i$. Then B is a Borel subset of M containing A. If we can justify for any smooth chart (U, φ) on M now that $\mu(\varphi(U \cap B)) = 0$ then we will be done.

Fortunately, note that:
$$\varphi(U \cap B) = \bigcup_{i \in \mathbb{N}} \varphi(U \cap U_i \cap B) = \bigcup_{i \in \mathbb{N}} (\varphi \circ \varphi_i^{-1})(\varphi_i(U \cap U_i \cap B)).$$

Then because $\varphi_i(U \cap U_i \cap B) \subseteq \varphi_i(U_i \cap B)$ and we know the latter is a Lebesgue null set, we know that $\varphi_i(U \cap U_i \cap B)$ is a Lebesgue null set. And by an application of proposition 6.5 from two pages ago plus the fact that the transition function $(\varphi \circ \varphi_i^{-1})$ is a C^r map, we can conclude that $(\varphi \circ \varphi_i^{-1})(\varphi_i(U \cap U_i \cap B))$ is also a Lebesgue null set. Since this works for all i, we can conclude that $\varphi(U \cap B)$ is a Lebesgue null set. ■

Proposition 6.8: If $A \subseteq M$ is a set of measure zero in M then $M - A$ is dense in M.

Proof:
If $M - A$ is not dense in M then we can find an open set U contained in A. But now let (V, ψ) be a smooth chart on M such that $V \cap U \neq \emptyset$. Then $\psi(V \cap U)$ is an open subset of $[0, \infty)^k \times \mathbb{R}^{n-k}$ (or $[0, \infty)^n$ or $\mathbb{R}^n$). It follows that $\psi(V \cap U)$ is measurable and will contain a coordinate ball, half ball, or corner ball. So, $\mu(\psi(V \cap U)) > 0$. And since $\mu(\psi(V \cap A)) \geq \mu(\psi(V \cap U))$ (assuming $\psi(V \cap A)$ is even measurable), we can conclude that A does not have measure zero in M. ■

Theorem 6.9: Suppose M and N are C^r n-dimensional manifolds and $F : N \to M$ is a C^1 map. If $A \subseteq M$ is a set of measure zero in M then $F(A)$ is a set of measure zero in N.

Proof:
Let $\{(U_k, \varphi_k)\}_{k \in \mathbb{N}}$ be a countable collection of smooth charts covering M. Then suppose (ψ, V) is any smooth chart on N. By lemma 6.6 it suffices to show that $\psi(V \cap F(A))$ is Lebesgue measurable with measure zero. Fortunately, note that:

$$\begin{aligned}\psi(V \cap F(A)) &= \psi(\textstyle\bigcup_{k \in \mathbb{N}}(V \cap F(A \cap U_k))) \\ &= \textstyle\bigcup_{k \in \mathbb{N}} \psi(V \cap F(A \cap U_k)) = \bigcup_{k \in \mathbb{N}}(\psi \circ F)(A \cap U_k \cap F^{-1}(V)) \\ &= \textstyle\bigcup_{k \in \mathbb{N}}(\psi \circ F \circ \varphi^{-1})(\varphi(A \cap U_k \cap F^{-1}(V))).\end{aligned}$$

Now because A has measure zero in M and $(U_k \cap F^{-1}(V), \varphi|_{F^{-1}(V)})$ is a smooth chart on M, we know from a prior lemma that $\varphi(A \cap U_k \cap F^{-1}(V))$ is Lebesgue measurable with measure zero for each k. In turn, because $\psi \circ F \circ \varphi^{-1}$ is a C^1 map, we can also say that $(\psi \circ F \circ \varphi^{-1})(\varphi(A \cap U_k \cap F^{-1}(V)))$ is Lebesgue measurable with measure zero for each k. And finally, since $\psi(V \cap F(A))$ is a countable union of these sets, we can conclude that $\psi(V \cap F(A))$ is Lebesgue measurable with measure zero. ■

9/16/2026

Today I want to cover Sard's theorem. This will be split into a few results. In particualar, I will first focus on the case of a function whose domain is an open subset of $\mathbb{R}^m$ and whose codomain is $\mathbb{R}^n$.

Lemma A: Suppose X is an LCH space, Y is a Hausdorff topological space, and $f : X \to Y$ is continuous.

- If X is second countable and $U \subseteq X$ is open then $f(U)$ is a Borel set.

  Proof:
  Note that if X is a second countable LCH space then U is a σ-compact set. As a result,

we can find a sequence of compact sets $\{K_\ell\}_{\ell\in\mathbb{N}}$ such that $U = \bigcup_{\ell\in\mathbb{N}} K_\ell$. But then $f(U) = \bigcup_{\ell\in\mathbb{N}} f(K_\ell)$. Since f is continuous and K_ℓ is compact, we know $f(K_\ell)$ is compact. In turn, as Y is Hausdorff we know that $f(K_\ell)$ is a closed set and thus Borel. Finally, $f(U)$ being a countable union of Borel sets means that $f(U)$ is a Borel set.

- If X is σ-compact and $C \subseteq X$ is closed then $f(C)$ is a Borel set.

  Proof:
  Since X is σ-compact, we can find a sequence of compact sets $\{K_\ell\}_{\ell\in\mathbb{N}}$ such that $X = \bigcup_{\ell\in\mathbb{N}} K_\ell$. Then $f(C) = \bigcup_{\ell\in\mathbb{N}} f(C \cap K_\ell)$. And by identical reasoning as before, we can conclude that $f(C \cap K_\ell)$ is a closed set for all ℓ. ■

Lemma B: Suppose $m < n$ are integers, $U \subseteq \mathbb{R}^m$ is an open set, and $f : U \to \mathbb{R}^n$ is a C^1 map. Then $f(U)$ is a Borel null set.

Proof:
By the prior lemma, we know that $f(U)$ is a Borel set. Also, we can define a C^1 map $g : U \times \mathbb{R}^{n-m} \to \mathbb{R}^n$ by $g(x_1, \ldots, x_n) = f(x_1, \ldots, x_k)$. But now $f(U) = g(U \times \{(0, \ldots, 0)\})$. Since $U \times \{(0, \ldots, 0)\}$ is a null set of $\mathbb{R}^n$, we can conclude by proposition 6.5 on *page 1139* that $g(U \times \{(0, \ldots, 0)\})$ has Lebesgue measure zero. ■

This is enough to to prove a trivial case of Sard's theorem.

Corollary (Sard Part 1): Suppose $m < n$, $U \subseteq \mathbb{R}^m$ is an open subset, and $f : U \to \mathbb{R}^n$ is a C^1 map. Then the set of critical values of f is a Lebesgue null set in $\mathbb{R}^n$.

Proof:
The set of critical values of f will be a subset of $f(U)$. But the prior lemma says that $f(U)$ is a Lebesgue null set. So, the critical values of f will also be a Lebesgue null set. ■

Next I need to cover the actual base case for the difficult version of Sard's Theorem. However, first I want to go on a tangent about vector valued integration.

In my math 140b notes I defined integration of an $\mathbb{R}^n$-valued function $f = (f_1, \ldots, f_n)$ by just taking the Riemann integral of f component wise. Meanwhile, on *pages 515-517* I defined a generalization of the Lebesgue integral for $\mathcal{Y}$-valued functions where $\mathcal{Y}$ is a separable Banach space. I'd like to try to merge those approaches.

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\mathcal{H}$ be a separable Hilbert space with an orthonormal basis $\{u_k\}_{k\in\mathbb{N}}$. If f is $\mathcal{H}$-valued then we will let $\int f \mathrm{d}\mu$ refer to the Bochner integral from *pages 515-517*.

Lemma C: Suppose X is a second countable and regular (i.e. T_3) topological space. Then X is normal (i.e. T_4).

Proof:
Let A and B be disjoint closed subsets of X. For every $x \in A$ we can use regularity to find an open set U_x containing x that is disjoint from B. Next, we can use regularity to find two disjoint open sets U'_x and $\widetilde{U'_x}$ such that $x \in U'_x$ and $(U_x)^C \subseteq \widetilde{U'_x}$. Finally, we can use

regularity one more time to find an open set $U_x'' \subseteq U_x'$ containing x. The result of doing this is that the closure of U_x'' is disjoint from B. After all, $B \subseteq (U_x)^C \subseteq \widetilde{U_x'}$. So, $(\widetilde{U_x'})^C$ is a closed set disjoint from B that contains U_x'. And since $U_x'' \subseteq U_x'$, we can conclude there is a closed set containing U_x'' that is disjoint from B. So, $\overline{U_x''} \cap B = \emptyset$.

By identical reasoning, for every $y \in B$ we can find an open set V_y'' containing y such that $\overline{V_y''} \cap A = \emptyset$.

But now since X is second countable, we know there are sequences $\{x_n\}_{n\in\mathbb{N}} \subseteq A$ and $\{y_n\}_{n\in\mathbb{N}} \subseteq B$ such that $A \subseteq \bigcup_{n\in\mathbb{N}} U_{x_n}''$ and $B \subseteq \bigcup_{n\in\mathbb{N}} V_{y_n}''$.

To finish off, let $\widehat{U}_n := U_{x_n} - \bigcup_{k=1}^n \overline{V_{y_k}''}$ and $\widehat{V}_n := V_{y_n} - \bigcup_{k=1}^n \overline{U_{x_k}''}$. Then for every n we have that $\widehat{U}_n$ and $\widehat{V}_n$ are open subsets of X. Furthermore, we still have that $A \subseteq \bigcup_{n\in\mathbb{N}} \widehat{U}_n$. After all, every $x \in A$ is contained in some U_{x_n} but is not contained in any of the $\overline{V_{y_n}}$. Similarly, we have that $B \subseteq \bigcup_{n\in\mathbb{N}} \widehat{V}_n$.

Let $U := \bigcup_{n\in\mathbb{N}} \widehat{U}_n$ and $V := \bigcup_{n\in\mathbb{N}} \widehat{V}_n$. Then U and V are open neighborhoods of A and B respectively. I also claim U and V are disjoint. After all, suppose $x \in U \cap V$. Then $x \in \widehat{U}_j \cap \widehat{V}_k$ for some j and k. By symmetry, we can assume without loss of generality that $j \leq k$. But now $x \in V_k'' - \bigcup_{j'=1}^k \overline{U_{j'}''}$ and $x \in U_j$. This is a contradiction. ■

Lemma D: Suppose X is a second countable and regular (i.e. T_3) topological space. Then given any closed subset $C \subseteq X$ we can find a decreasing sequence of open sets $\{V_k\}_{k\in\mathbb{N}}$ in X such that $\overline{V_{k+1}} \subseteq V_k$ for all $k \in \mathbb{N}$ and $C = \bigcap_{k\in\mathbb{N}} V_k$.

Proof:
If $C = X$ then this lemma is trivial. Just set $V_k = X$ for all k. So from now on we'll suppose C is a proper subset of X.

Consider any $x \in C$. By regularity we can find open sets U_x and N_x such that $C \subseteq U_x$ and $x \in N_x$, and $U_x \subseteq X - N_x$. But then because X is second countable, we can find a sequence $\{x_k\}_{k\in\mathbb{N}} \subseteq X - C$ such that $X - C = \bigcup_{k\in\mathbb{N}} N_{x_k}$. In turn, we have that $C = \bigcap_{k\in\mathbb{N}}(X - N_{x_k}) \supseteq \bigcap_{k\in\mathbb{N}}(U_{x_k}) \supseteq C$. And now we set $W_k := \bigcap_{j=1}^k U_{x_j}$ for all $k \in \mathbb{N}$. By doing this we get a decreasing sequence of open sets $\{W_k\}_{k\in\mathbb{N}}$ in X such that $C = \bigcap_{k\in\mathbb{N}} W_k$.

Next, we want to modify our new sets to get the desired closure property. Fortunately, by lemma C we know that X is a normal (i.e. T_4) topological space. As a result, given any open set U containing C we can easily find an open set W such that $C \subseteq W \subseteq \overline{W} \subseteq U$.
Namely, by Urysohn's lemma we can find a function $f : X \to [0,1]$ such that $C \subseteq f^{-1}(\{0\})$ and $U^C \subseteq f^{-1}(\{1\})$ Then just set $W = f^{-1}([0, \frac{1}{2}))$.

Set $V_1 = W_1$. Then for every $k > 1$ we inductively define V_k as follows.
Let Ω_k be an open subset of X such that $C \subseteq \Omega_k \subseteq \overline{\Omega_k} \subseteq V_{k-1}$. Then set $V_k := \Omega_k \cap W_k$. Thus $\overline{V_k} \subseteq \overline{\Omega_k} \subseteq V_{k-1}$. Also $V_k \subseteq W_k$. And since C is contained in W_k and Ω_k, we have that $C \subseteq V_k$.

Because $C = \bigcap_{k\in\mathbb{N}} W_k$, and $C \subseteq V_k \subseteq W_k$ for all k, we can conclude that $C = \bigcap_{k\in\mathbb{N}} V_k$ still. ■

Lemma E: Suppose $(X, \mathcal{M})$ is a measurable space, and Y is a regular (i.e T_3) second countable topological space equipped with it's Borel σ-algebra, and $\{f_n : X \to Y\}_{n\in\mathbb{N}}$ is a sequence of functions converging pointwise to $f : X \to Y$. If every f_n is $(\mathcal{M}, \mathcal{B}_Y)$-measurable then so is f.

Proof:
Since the closed subsets of Y generate the Borel σ-algebra, it suffices to show that $f^{-1}(C) \in \mathcal{M}$ for any arbitrary closed sets $C \subseteq Y$. Fortunately, by the prior lemma we can find a decreasing sequence of open sets $\{V_k\}_{k\in\mathbb{N}}$ such that $C = \bigcap_{k\in\mathbb{N}} V_k$ and $\overline{V_{k+1}} \subseteq V_k$ for all $k \in \mathbb{N}$.

Next note that if $f(x) \in C$ then we know by the definition of a pointwise limit that for every set V_k there exists an integer $N \in \mathbb{N}$ such that $f_n(x) \in V_k$ for all $n \geq N$. In other words:

$$f^{-1}(C) \subseteq \bigcap_{k\in\mathbb{N}} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} f_n^{-1}(V_k)$$

On the other hand, if $f(x) \notin C$ then because $C = \bigcap_{k\in\mathbb{N}} \overline{V_k}$ we know that $Y - C = \bigcup_{k\in\mathbb{N}}(Y - \overline{V_k})$ Hence, there exists some $k \in \mathbb{N}$ such that $f(x) \notin \overline{V_k}$. Next, because $Y - \overline{V_k}$ is an open set, we know by the definition of a limit that there exists an integer $N \in \mathbb{N}$ such that $f_n(x) \in X - \overline{V_k}$ for all $n \geq N$. Consequently, for all $N \in \mathbb{N}$ we have that $x \notin \bigcap_{n\geq N} f_n^{-1}(V_k)$. So, $x \notin \bigcup_{N\in\mathbb{N}}(\bigcap_{n\geq N} f_n^{-1}(V_k))$. And this proves that:

$$f^{-1}(C) \supseteq \bigcap_{k\in\mathbb{N}} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} f_n^{-1}(V_k)$$

So, now that we know that $f^{-1}(C) = \bigcap_{k\in\mathbb{N}}(\bigcup_{N\in\mathbb{N}}(\bigcap_{n\geq N} f_n^{-1}(V_k)))$, we can conclude that $f^{-1}(C) \in \mathcal{M}$. After all, f_n being measurable and V_k being open means that $f_n^{-1}(V_k)$ is open for all k and n. Plus, $\mathcal{M}$ is closed under countable unions and intersections. ■

In particular, if $\mathcal{H}$ is a separable Hilbert space, then $\mathcal{H}$ will be second countable and normal (since every metric space is normal and every separable metric space is second countable). So, lemma E applies to $\mathcal{H}$.

Proposition: Let $L_{\mathcal{H}}$ be the space of all $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable maps from X to $\mathcal{H}$ (where $\mathcal{B}_{\mathcal{H}}$ is the Borel σ-algebra on $\mathcal{H}$). Then $f : X \to \mathcal{H}$ is in $L_{\mathcal{H}}$ if and only if $f_k(x) := \langle f(x), u_k\rangle$ is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable for all $k \in \mathbb{N}$.

$(\Longrightarrow)$
Let $\pi_k : \mathcal{H} \to \mathbb{C}$ be the map $\pi_k(x) = \langle x, u_k\rangle$. This is a continuous linear functional on $\mathcal{H}$ and so it is trivially $(\mathcal{B}_{\mathcal{H}}, \mathcal{B}_{\mathbb{C}})$-measurable. Also note that $f_k = \pi_k \circ f$. So if f is $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable then we can conclude that f_k is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable.

$(\Longleftarrow)$
Because $\{u_k\}_{k\in\mathbb{N}}$ is a (complete) orthonormal basis for $\mathcal{H}$, we know that:

$$f(x) = \textstyle\sum_{k\in\mathbb{N}} \langle f(x), u_k\rangle u_k = \sum_{k\in\mathbb{N}} f_k(x) u_k.$$

But now if $m : \mathbb{C} \times \mathcal{H} \to \mathcal{H}$ is the scalar multiplication function, we know that m is continuous and in turn m is $(\mathcal{B}_{\mathbb{C}\times\mathcal{H}}, \mathcal{B}_{\mathcal{H}})$-measurable. But by a propositon at the top of page 14 in my math 240a notes as well as the fact that $\mathbb{C}$ and $\mathcal{H}$ are separable metric spaces, we know that $\mathcal{B}_{\mathbb{C}\times\mathcal{H}} = \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathcal{H}}$. Also, $g_k(x) = (f_k(x), u_k)$ is going to be a $(\mathcal{M}, \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathcal{H}})$-measurable function for every k. So by composing functions, we get that $x \mapsto f_k(x)u_k$ is a $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable function for all k.

To finish off, since finite sums of measurable functions from X to $\mathcal{H}$ are always measurable, we can just apply lemma E to get that f is $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable. ■

Proposition: Suppose $f \in L^1_{\mathcal{H}} := \{g \in L_{\mathcal{H}} : \int \|g(x)\| \mathrm{d}\mu(x) < \infty\}$. Then $f_k(x) := \langle f(x), u_k \rangle \in L^1(X)$ and $\int f \mathrm{d}\mu = \sum_{k\in\mathbb{N}} (\int f_k \mathrm{d}\mu) u_k$.

Proof:

Because of the prior proposition we know that f_k is measurable for all k. Also note $|f_k(x)| \leq \|u_k\| \cdot \|f(x)\| = \|f(x)\|$. Therefore $\int |f_k| \mathrm{d}\mu \leq \int \|f\| \mathrm{d}\mu < \infty$ and this proves that $f_k \in L^1(X)$ for all $k \in \mathbb{N}$.

Next note that by reusing the reasoning in the prior proposition, we know that the map $x \mapsto f_k(x)u_k$ is a $(\mathcal{M}, \mathcal{B}_{\mathcal{H}})$-measurable map for all k. In turn the partial sums of $f(x) = \sum_{k=1}^{\infty} f_k(x)u_k$ are also measurable. Also note that:

$$\|\sum_{k=1}^{n} f_k(x)u_k\| = (\sum_{k=1}^{n} |f_k(x)|^2)^{1/2} \leq (\sum_{k=0}^{\infty} |f_k(x)|^2)^{1/2} = \|f(x)\|$$

So, $\int \|\sum_{k=1}^{n} f_k(x)u_k\| \mathrm{d}\mu \leq \int \|f(x)\| \mathrm{d}\mu < \infty$, meaning all our partial sums are in $L^1_{\mathcal{H}}$. And now we can use the dominated convergence theorem (with an upper bound of $g(x) = \|f(x)\|$) to say that:

$$\int f(x) \mathrm{d}\mu = \lim_{n\to\infty} \int \sum_{k=1}^{n} f_k(x)u_k \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} \int f_k(x)u_k \mathrm{d}\mu.$$

By considering the continuous linear map $T : \mathbb{C} \to \mathcal{H}$ given by $c_k \mapsto c_k u_k$, we know from *page 517* that $(\int f_k \mathrm{d}\mu)u_k = T(\int f_k \mathrm{d}\mu) = \int (T \circ f_k) \mathrm{d}\mu = \int f_k(x)u_k \mathrm{d}\mu$. So:

$$\int f(x) \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} \int f_k(x)u_k \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} (\int f_k \mathrm{d}\mu)u_k = \sum_{k=1}^{\infty} (\int f_k \mathrm{d}\mu)u_k. \blacksquare$$

For the next bit of my notes I'm learning from a different pdf on the internet (see *citation 13*).

Theorem (Sard Part 2): Suppose $U \subseteq \mathbb{R}^n$ is an open set and $f : U \to \mathbb{R}^n$ is a C^1 map. Then the set of critical values of f is a Borel null set in $\mathbb{R}^n$.

Proof:

Recall from proposition 4.1 on *Pages 1054-1056* that the set of regular points of f is open. It follows that the set C of critical points of f is closed (relative to U). And since $f(C)$ is the set of critical values of f, we can conclude from lemma A that $f(C)$ is a Borel subset of $\mathbb{R}^n$.

Next note that since U is second countable, we can cover U with countably many closed

coordinate cubes. In other words, for all $k \in \mathbb{N}$ we can find $x_k = (x_k^1, \ldots, x_k^n) \in U$ and $\varepsilon_k > 0$ such that $U = \bigcup_{k\in\mathbb{N}}(\prod_{i=1}^n [x_k^i - \varepsilon, x_k^i + \varepsilon])$. And now upon setting $E_k := \prod_{i=1}^n [x_k^i - \varepsilon, x_k^i + \varepsilon]$, it suffices to prove that $f(C \cap E_k)$ is a Borel null set for all k.

As a side note, $f(C \cap E_k)$ will be Borel measurable for identical reasons that $f(C)$ is Borel measurable.

Fortunately, note that by applying a sufficient affine transformation to the domain of f, we can without loss of generality assume $E_k = [0,1]^n =: I^n$. After all, an affine transformation is a C^∞ diffeomorphism and will thus not effect the critical values of f at all. So, we are now left with the slightly simpler task of proving that if $f : I^n \to \mathbb{R}^n$ is a C^1 map and $C \subseteq I^n$ is the set of critical points of f then $f(C)$ is a Borel null set. Two advantages of our simplified task though are as follows.

1. I^n a compact subset of $\mathbb{R}^n$.
2. By proposition C.29 on *page 291*, we know f is Lipschitz on I^n. So let $M \geq 0$ be some constant such that $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in I^n$.

Next note for any $x, c \in I^n$ that:

$$\begin{aligned} f(x) - f(c) - (\mathrm{D}f(c))(x-c) &= \textstyle\int_0^1 (\mathrm{D}f(c + t(x-c)))(x-c)\mathrm{d}t - (\mathrm{D}f(c))(x-c) \\ &= \textstyle\int_0^1 [\mathrm{D}f(c + t(x-c)) - \mathrm{D}f(c)](x-c)\mathrm{d}t. \end{aligned}$$

In turn we have that:

$$\begin{aligned} |f(x) - f(c) - (\mathrm{D}f(c))(x-c)| &\leq \left(\textstyle\int_0^1 \|\mathrm{D}f(c + t(x-c)) - \mathrm{D}f(c)\|_{\mathrm{op}}\mathrm{d}t\right)|x-c| \\ &\leq \left(\sup_{x' \in B_{|x-c|}(c)} \|\mathrm{D}f(x') - \mathrm{D}f(c)\|_{\mathrm{op}}\right) \cdot |x-c| \end{aligned}$$

Then as f is a C^1 map, we know that $x \mapsto \mathrm{D}f(x)$ is continuous as a map from I^n to $B(\mathbb{R}^n)$. Next, since I^n is compact we know that $x \mapsto \mathrm{D}f(x)$ is uniformly continuous as a map from I^n to $B(\mathbb{R}^n)$. So, for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|x' - c| < \delta$ implies that $\|\mathrm{D}(x') - \mathrm{D}(c)\|_{\mathrm{op}} < \varepsilon$. So in particular, for all $\varepsilon > 0$ we there exists $\delta > 0$ such that for all $x, c \in I^n$ with $|x - c| < \delta$ we have that $|f(x) - f(c) - (\mathrm{D}f(c))(x-c)| \leq \varepsilon|x-c|$.

Yet now suppose $c \in C$. Then $\mathrm{im}(\mathrm{D}f(c))$ is contained in an $(n-1)$-dimensional hyperplane H. And if $|x - c| < \kappa < \delta$ then:

$$\inf_{x' \in H} |f(x) - f(x')| \leq |f(x) - f(c) - \mathrm{D}f(c)(x-c)| \leq \varepsilon\kappa.$$

Similarly, $|f(x) - f(c)| \leq M\kappa$. So, all in all we can say that there is some n-dimensional parallelotope with volume $(2M\kappa)^{n-1} \cdot (2\varepsilon\kappa) = 2^n \kappa^n M^{n-1} \cdot \varepsilon$ containing $f(B_\kappa(c) \cap I^n)$.

To finish off, we can subdivide I^n into h^n many subcubes of side length h^{-1} (where h is a sufficiently large integer such that $h^{-1}\sqrt{n} < \delta$). Note that the side-lengths of any subcube (say Q) are small enough to guarentee that if $c \in Q \cap C$ then $Q \subseteq B_{h^{-1}\sqrt{n}}(c)$. So in turn, by the prior reasoning we'd have that $f(Q) \subseteq f(B_{h^{-1}\sqrt{n}}(c) \cap I^n)$ where the latter is contained in a parallelotope with volume $2^n h^{-n} n^{n/2} M^{n-1} \cdot \varepsilon$. There are only h^n many subcubes though so at most h^n of them actually intersect C. Hence, by just taking the unions of the images of the subcubes that intersect C, we can guarentee that $f(C)$ is contained in a set of measure at most $h^n(2^n h^{-n} n^{n/2} M^{n-1} \cdot \varepsilon) = 2^n n^{n/2} M^{n-1} \cdot \varepsilon$.

But ε can be any positive real number. So, we just take $\varepsilon \to 0$ to get that $\mu(f(C)) = 0$. ■

Finally, note that if $m > n$, $U \subseteq \mathbb{R}^m$ is an open set, and $f : U \to \mathbb{R}^n$ is a C^1 map, then Sard's theorem as was priorly stated does not hold anymore. I don't have the background to explicitely construct a counter example, but here's some vague reasoning for why we wouldn't expect Sard's theorem to hold.

Because the Cantor set $C \subseteq \mathbb{R}$ is a null set, it wouldn't contradict the prior theorem that there exists a C^1 map $f : \mathbb{R} \to \mathbb{R}$ such that the set of critical values of f includes C. (As it turns out, such a function does exist although I don't know how to construct it.)

Next define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x, y) = f(x) + f(y)$. Then g is a C^1 map. But also note that $C + C$ is a subset of the critical values of f. And we can fairly easily conclude that $C + C = [0, 2]$.

Why?
Since $C \subseteq [0, 1]$ we know that $C+C \subseteq [0, 2]$. On the other hand, suppose $x \in [0, 2]$ and let $y = \frac{x}{2}$. We write the base 3 representation of y as $\sum_{k=1}^{\infty} a_k \frac{1}{3^k}$. Now if $a_k = 0$ we write $b_k = c_k = 0$. If $a_k = 1$ we write $b_k = 1$ and $c_k = 0$. And if $a_k = 2$ we write $b_k = c_k = 1$. Then upon setting $z_1 = \sum_{k=1}^{\infty} b_k \frac{1}{3^k}$ and $z_2 = \sum_{k=1}^{\infty} c_k \frac{1}{3^k}$, we have that $y = z_1 + z_2$. In turn, $x = 2z_1 + 2z_2$. Also $2z_1, 2z_2 \in C$ since all their ternary-digits are 0 and 2. ■

So, the set of critical values of g would have positive measure.

As it turns out we can get Sard's theorem to still work though by requiring f to have more derivatives.

Theorem (Sard Part 3): Suppose $m > n$, $U \subseteq \mathbb{R}^m$ is an open subset, and $f : U \to \mathbb{R}^n$ is a C^k map where $k \geq \frac{m}{n}$. Then the set of critical values of f is a Borel measurable set in $\mathbb{R}^n$ with Lebesgue measure 0.

Proof:
We'll do induction on m, noting that Sard parts 1 and 2 already proved the base case of when $m = 1$. Also, by identical reasons as with Sard part 2, it suffices to assume that the domain of f is actually the compact set $I^m := [0, 1]^m$.

Let $C_j := \{x \in I^m : \partial^\alpha f(x) = 0$ for all multi-indices α such that $|\alpha| \leq j\}$. Also let C be the set of critical points of f. We know C and all the C_j are closed subsets of $\mathbb{R}^n$ points (either because the set of critical values is always closed or because the preimage under a continuous function of a closed set is closed). Also $C \supseteq C_1 \supseteq \cdots \supseteq C_k$.

Note that $C - C_1$ and $C_i - C_{i+1}$ are open subsets of C or C_i respectively where both of those are second countable (due to being subspaces of a second countable space) and locally compact Hausdorff (due to being closed subspace of a LCH space). So by lemma A on *pages 1141-1142* we can conclude that $f(C-C_1)$ and $f(C_i-C_{i+1})$ are Borel measurable for all $i < k$. Also because C_k is a closed subset of I^n, we can similarly conclude that $f(C_k)$ is Borel measurable.

Now clearly $f(C) = f(C_k) \cup (f(C) - f(C_1)) \cup \bigcup_{i=1}^{k-1}(f(C_i) - f(C_{i+1}))$. Our strategy will be to show that each individual set in this union is a null set.

- Given any multidex α and any $x \in \mathbb{R}^m$, recall from my math 240c notes how ∂^α, x^α, $\alpha!$, and $|\alpha|$ are defined.

Lemma (Another Version of Taylor's Theorem): Suppose $U \subseteq \mathbb{R}^n$ is open and $g \in C^{k+1}(U, \mathbb{R})$. Then for any $c \in U$ and convex subset $W \subseteq U$ containing c, we have for all $x \in W$ that:

$$g(x) = \sum_{|\alpha| \leq k} \frac{\partial^\alpha g(c)}{\alpha!}(x-c)^\alpha + R_k(x)$$

where $R_k(x) = \sum_{|\alpha|=k+1} \frac{k+1}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^k \partial^\alpha g(c + t(x-c))\mathrm{d}t$.

Proof:
Let $S_\ell := \{1, \ldots, n\}^\ell$. Also let us write $x = (x_1, \ldots, x_n)$ and $c = (c_1, \ldots, c_n)$. Then I'm going to prove that $g(x) = \widetilde{P}_k(x) + \widetilde{R}_k(x)$ where:

- $\widetilde{P}_k(x) := g(c) + \sum_{\ell=1}^{k} \left(\frac{1}{\ell!} \sum_{(i_1,\ldots,i_\ell) \in S_\ell} \left(\prod_{j=1}^{\ell} (x_{i_j} - c_{i_j}) \right) \frac{\partial^\ell g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_\ell}}}(c) \right)$,
- $\widetilde{R}_k(x) := \frac{1}{k!} \sum_{(i_1,\ldots,i_{k+1}) \in S_{k+1}} \left(\prod_{j=1}^{k+1} (x_{i_j} - c_{i_j}) \right) \int_0^1 (1-t)^k \frac{\partial^{k+1} g}{\partial x_{i_1} \cdots \partial_{x_{i_{k+1}}}}(c + t(x-c))\mathrm{d}t$.

Note that if $|\alpha| = \ell$ then there are $\frac{\ell!}{\alpha!}$ many orders one can differentiate in to get $\partial^\alpha g$. So by collecting together like terms, we can see that $\widetilde{R}_k = R_k$ and $\widetilde{P}_k = P_k$.

To start off, note that if $k = 0$ then I've already shown this result on *page 1035*.
So now we proceed by induction on k, supposing we've already shown that $g(x) = \widetilde{P}_{k-1}(x) + \widetilde{R}_{k-1}(x)$

We get via integration by parts that:

$$\begin{aligned}
&\int_0^1 (1-t)^{k-1} \frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c + t(x-c))\mathrm{d}t \\
&= \left[\frac{-(1-t)^k}{k} \frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c + t(x-c)) \right]_{t=0}^{t=1} + \int_0^1 \frac{(1-t)^k}{k} \cdot \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c + s(x-c)) \right](t)\mathrm{d}t \\
&= \frac{1}{k} \frac{\partial^k g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}}}(c) + \sum_{j=1}^{n} \frac{(x_j - c_j)}{k} \int_0^1 (1-t)^k \frac{\partial^{k+1} g}{\partial_{x_{i_1}} \cdots \partial_{x_{i_k}} \partial_{x_j}}(c + t(x-c))\mathrm{d}t
\end{aligned}$$

And now we shift the term without the integral into being part of $\widetilde{P_k}$ and we let the term with the integral be in $\widetilde{R}_k$. ■

Corollary: Suppose $U \subseteq \mathbb{R}^n$ is open and $g \in C^{k+1}(U, \mathbb{R})$. Given any $c \in U$ let P_k and R_k be from the prior lemma. Then suppose $W \subseteq U$ is a convex subset such that $c \in W$ and all $(k+1)$th partial derivatives of g are bounded in absolute value on W by a constant M. Then for all $x \in W$ we have that:

$$|f(x) - P_k(x)| = |R_k(x)| \leq \frac{n^{k+1} M}{(k+1)!}|x-c|^{k+1}$$

Proof:
Note that:

$$\frac{k+1}{\alpha!}(x-c)^\alpha \int_0^1 (1-t)^k \partial^\alpha g(c+t(x-c))\mathrm{d}t \leq \frac{M(k+1)}{\alpha!}|x-c|^{|\alpha|}\int_0^1(1-t)^k\mathrm{d}t$$
$$\leq \frac{M}{\alpha!}|x-c|^{k+1}$$

So, we can say that $|R_k(x)| \leq M|x-c|^{k+1}\sum_{|\alpha|=k+1}\frac{1}{\alpha!}$.

Also note that: $\sum_{|\alpha|=k+1}\frac{1}{\alpha!} = \frac{1}{(k+1)!}\sum_{|\alpha|=k+1}\frac{(k+1)!}{\alpha!} = \frac{n^{k+1}}{(k+1)!}$. ■

Corollary 2: Suppose $U \subseteq \mathbb{R}^m$ is open and $g \in C^k(U,\mathbb{R}^n)$. Also suppose $K \subseteq U$ is a compact convex subset. Let $P_k(x,c) = \sum_{|\alpha|\leq k}\frac{(x-c)^\alpha}{\alpha!}\partial^\alpha g(c)$. Then for any $\varepsilon > 0$ we can find $\delta > 0$ such that $|x-c| < \delta \implies |g(x) - P(x,c)| \leq M\varepsilon|x-c|^k$. (Here M does not depend on ε and δ.)

Proof:

Write $g(x) = (g_1(x),\ldots,g_n(x))$. By the prior lemma, we know that:

$$g_i(x) - \sum_{|\alpha|<k}\frac{\partial^\alpha g_i(c)}{\alpha!}(x-c)^\alpha = \sum_{|\alpha|=k}\frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\partial^\alpha g_i(c+t(x-c))\mathrm{d}t$$

But now note:

$$\frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\partial^\alpha g_i(c+t(x-c))\mathrm{d}t - \frac{1}{\alpha!}(x-c)^\alpha\partial^\alpha g_i(c)$$
$$= \frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\partial^\alpha g_i(c+t(x-c))\mathrm{d}t - \frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\partial^\alpha g_i(c)\mathrm{d}t$$
$$= \frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\left[\partial^\alpha g_i(c+t(x-c)) - \partial^\alpha g_i(c)\right]\mathrm{d}t$$

So by subtracting all needed terms from both sides, we can get that:

$$g_i(x) - \textstyle\sum_{|\alpha|\leq k}\frac{(x-c)^\alpha}{\alpha!}\partial^\alpha g_i(c) = \sum_{|\alpha|=k}\frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\left[\partial^\alpha g_i(c+t(x-c)) - \partial^\alpha g_i(c)\right]\mathrm{d}t.$$

And by putting all the vector components together, we get that:

$$g(x) - P(x,c) = \sum_{|\alpha|=k}\frac{k}{\alpha!}(x-c)^\alpha\int_0^1(1-t)^{k-1}\left[\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c)\right]\mathrm{d}t.$$

But now since $\partial^\alpha g$ is a continuous function on K for all $|\alpha| = k$, we can guarentee that each of them are uniformly continuous. So, given any $\varepsilon > 0$ we can find $0 < \delta < \varepsilon$ such that $|x-c| < \delta \implies |\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c)| < \varepsilon$ for all multi-indices α with $|\alpha| = k$ and all $t \in [0,1]$. And now:

$$|g(x) - P(x,c)| \leq \sum_{|\alpha|=k}\frac{k}{\alpha!}|(x-c)^\alpha|\cdot\left|\int_0^1(1-t)^{k-1}\left[\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c)\right]\mathrm{d}t\right|$$
$$\leq \sum_{|\alpha|=k}\frac{k}{\alpha!}|x-c|^k\cdot\int_0^1(1-t)^{k-1}\left|\partial^\alpha g(c+t(x-c)) - \partial^\alpha g(c)\right|\mathrm{d}t$$
$$\leq \sum_{|\alpha|=k}\frac{k}{\alpha!}|x-c|^k\cdot\varepsilon\int_0^1(1-t)^{k-1}\mathrm{d}t = \sum_{|\alpha|=k}\frac{\varepsilon}{\alpha!}|x-c|^k = \frac{m^k\varepsilon}{k!}|x-c|^k$$

So upon setting $M = \frac{m^k}{k!}$ we have that $|g(x) - P(x,c)| \leq M\varepsilon|x-c|^k$.

By the prior corollary, for any $\varepsilon > 0$ we can find some $0 < \delta < \varepsilon$ such that for all $c \in C_k$ and $x \in I^m$, $|x-c| < \delta \implies |f(x) - f(c)| \leq M\varepsilon|x-c|^k$ (where $M = \frac{m^k}{k!}$ is independent of ε).

But now subdivide I^m into h^m many subcubes with sidelength h^{-1} where h is an integer with $h^{-1}\sqrt{m} < \delta$. Then for any subcube Q, can guarentee that if $c \in Q \cap C$ then $Q \subseteq B_{h^{-1}\sqrt{m}}(c)$. So by the prior line, this in turn guarentees that $f(Q) \subseteq f(B_{h^{-1}\sqrt{m}}(c) \cap I^m) \subseteq B_{M\varepsilon h^{-k} m^{k/2}}(f(c))$. So $f(Q)$ is contained in a set of measure $(M\varepsilon h^{-k} m^{k/2})^n \cdot \mu(B_1(0))$.

Since there are h^m many possible cubes that can intercept C_k, we can guarentee that:

$$\mu(f(C_k)) \leq h^m (M\varepsilon h^{-k} m^{k/2})^n \cdot \mu(B_1(0)) = \varepsilon^n \cdot M m^{kn/2} \mu(B_1(0)) h^{m-kn}$$

But now as $k \geq \frac{m}{n}$, we know that $m - kn \leq 0$. And since $h \geq 1$, we can thus conclude that $\mu(f(C_k)) \leq \varepsilon^n \cdot M m^{kn/2} \mu(B_1(0))$. Finally, by taking $\varepsilon \to 0$ we can thus conclude that $\mu(f(C_k)) = 0$.

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